1 Video 1

★Transform Analysis of LTI Systems and Filter Design

Outline :

- Frequency phase response and group delay
- Minimum phase and all-pass systems
- Linear phase and FIR filter types
- Filter Design:
  - ↳ IIR : by bilinear transform from CT filters.
  - ↳ FIR : by windowing the ideal filter.

2 Video 2

★Phase Response and Group Delay

$H(e^{j\omega}) = |H(e^{j\omega})| \cdot \angle H(e^{j\omega})$ ⇒ magnitude and phase continuous functions of ω

↓

time shift , distortion

group delay : $\tau(\omega) = grd[H(e^{j\omega})] \triangleq -\frac{d}{d\omega}\{\angle H(e^{j\omega})\}$

↓

baş örnek

$\tau(\omega)$ → pozitif (delay)

↳ negatif (anti-causal)

Örnek-1 : Effect of Group delay (Başlık 5.1.2)

$x[n] = x_3[n] + x_1[n-M-1] + x_2[n-2M-2]$, $M = 60$

$x_1[n] = w[n] \cdot \cos(0.2\pi n)$

$x_2[n] = w[n] \cdot \cos(0.4\pi n - \frac{\pi}{2})$

$x_3[n] = w[n] \cdot \cos(0.8\pi n + \pi/5)$

$w[n] = \begin{cases} 0.54 - 0.46 \cdot \cos\left(\frac{2\pi n}{M}\right) & 0 \le n \le M \\ 0 & \text{otherwise} \end{cases}$

↓

hamming window

MATLAB

Editor - C:\Users\msi\Documents\MATLAB\lab3_2019\groupdelay.m

SystemFunction.m | groupdelay.m

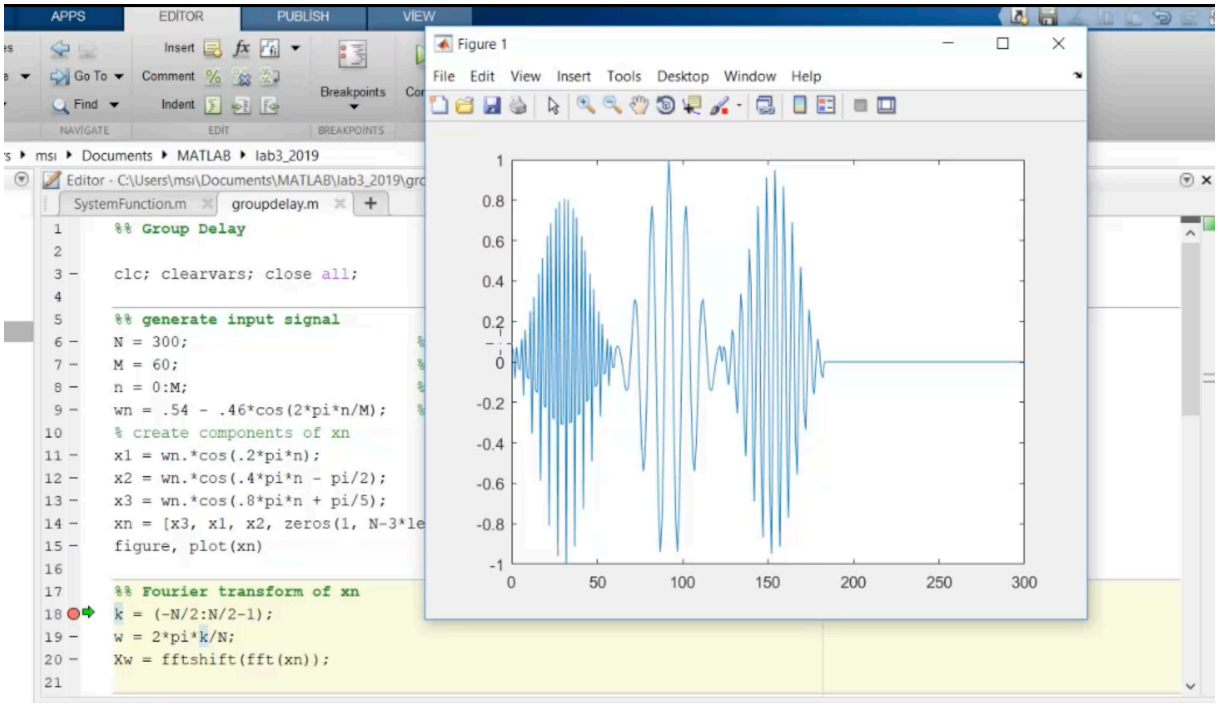
```
%% Group Delay

clc; clearvars; close all;

%% generate input signal
N = 300;                        % samples
M = 60;                         % size of the window
n = 0:M;                        % sample vector (time vector)
wn = .54 - .46*cos(2*pi*n/M);   % window (Hamming)
% create components of xn
x1 = wn.*cos(.2*pi*n);
x2 = wn.*cos(.4*pi*n - pi/2);
x3 = wn.*cos(.8*pi*n + pi/5);
xn = [x3, x1, x2, zeros(1, N-3*length(n))]; % input signal
figure, plot(xn)

%% Fourier transform of xn
k = (-N/2:N/2-1);
w = 2*pi*k/N;
Xw = fftshift(fft(xn));
```

Command Window




```
%% Fourier transform of xn
k = (-N/2:N/2-1);
w = 2*pi*k/N;
Xw = fftshift(fft(xn));

%% Generate original system
[Hw, p, z, K] = SystemFunction(w);
figure,
subplot 211, plot(w/pi, abs(Hw)), title('Frequency Response of H(w)'), grid on
subplot 212, plot(w/pi, (angle(Hw))), grid on
```

Editor - C:\Users\msi\Documents\MATLAB\lab3_2019\SystemFunction.m

SystemFunction.m | groupdelay.m

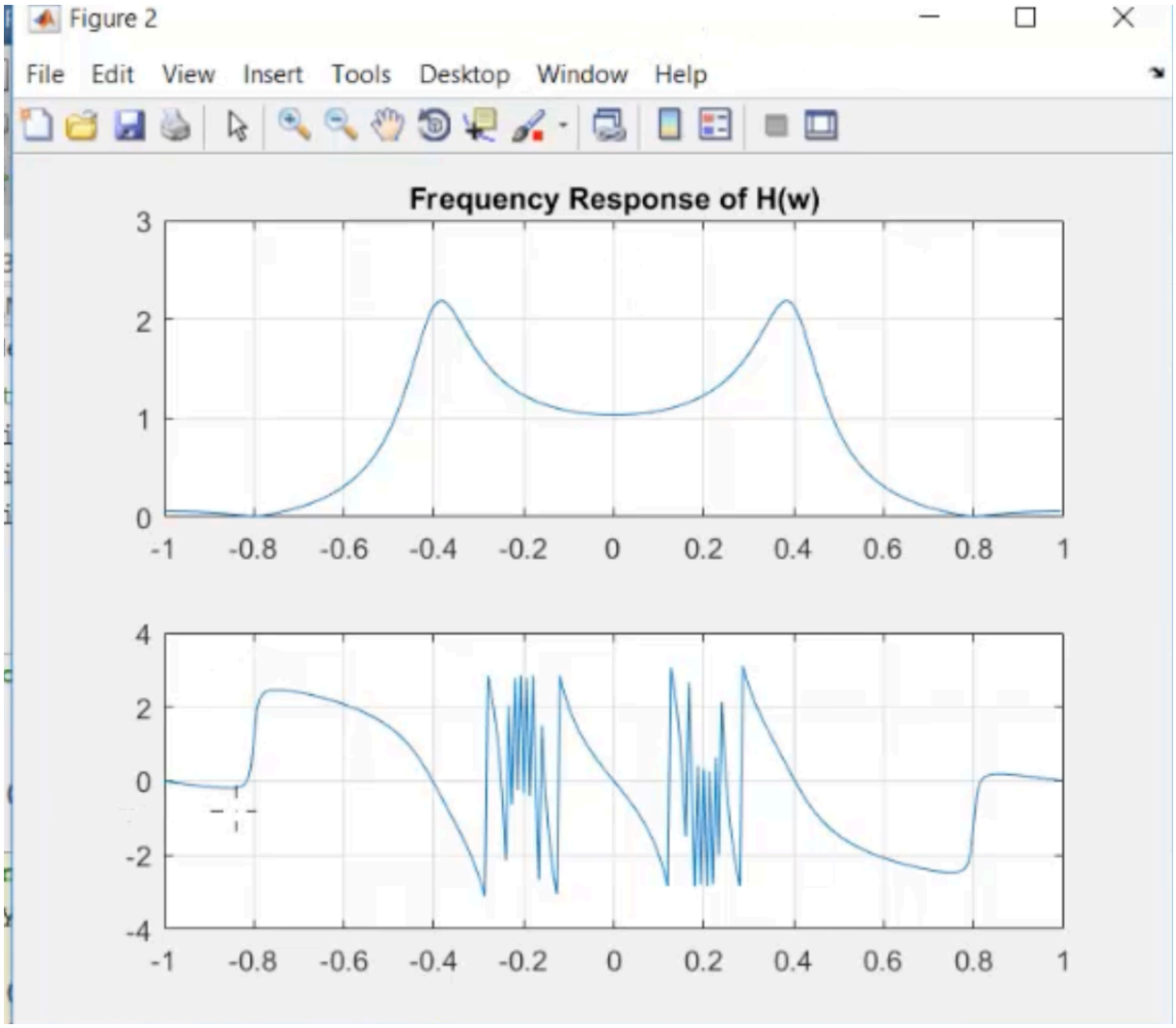
```
function [Hw, polezz, zerozz, K] = SystemFunction(w)

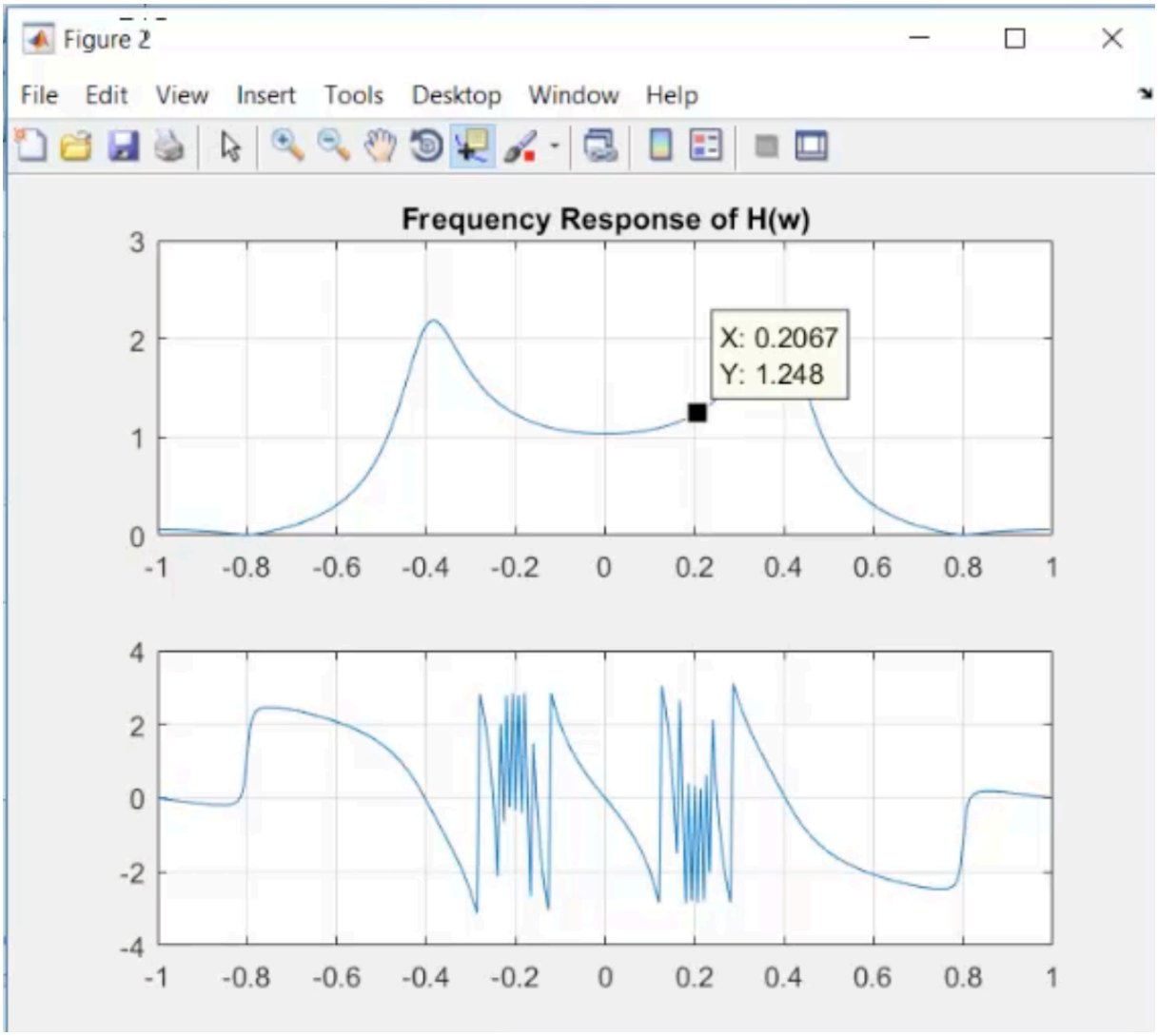
kk = 1:4;
ck = 0.95*exp(1j*(0.15*pi+0.02*pi*kk));
z1 = exp(-1j*w);
zs = [.98*exp(1j*.8*pi), .98*exp(-1j*.8*pi)];
ps = [.8*exp(1j*.4*pi), .8*exp(-1j*.4*pi)];
Hw = (1-zs(1)*z1).*(1-zs(2)*z1) ./ ((1-ps(1)*z1).*(1-ps(2)*z1));
for k = 1:length(kk)
    dummy = (conj(ck(k))-z1).*(ck(k)-z1) ./ ((1-ck(k)*z1).*(1-conj(ck(k))*z1));
    Hw = Hw .* dummy.^2;
end

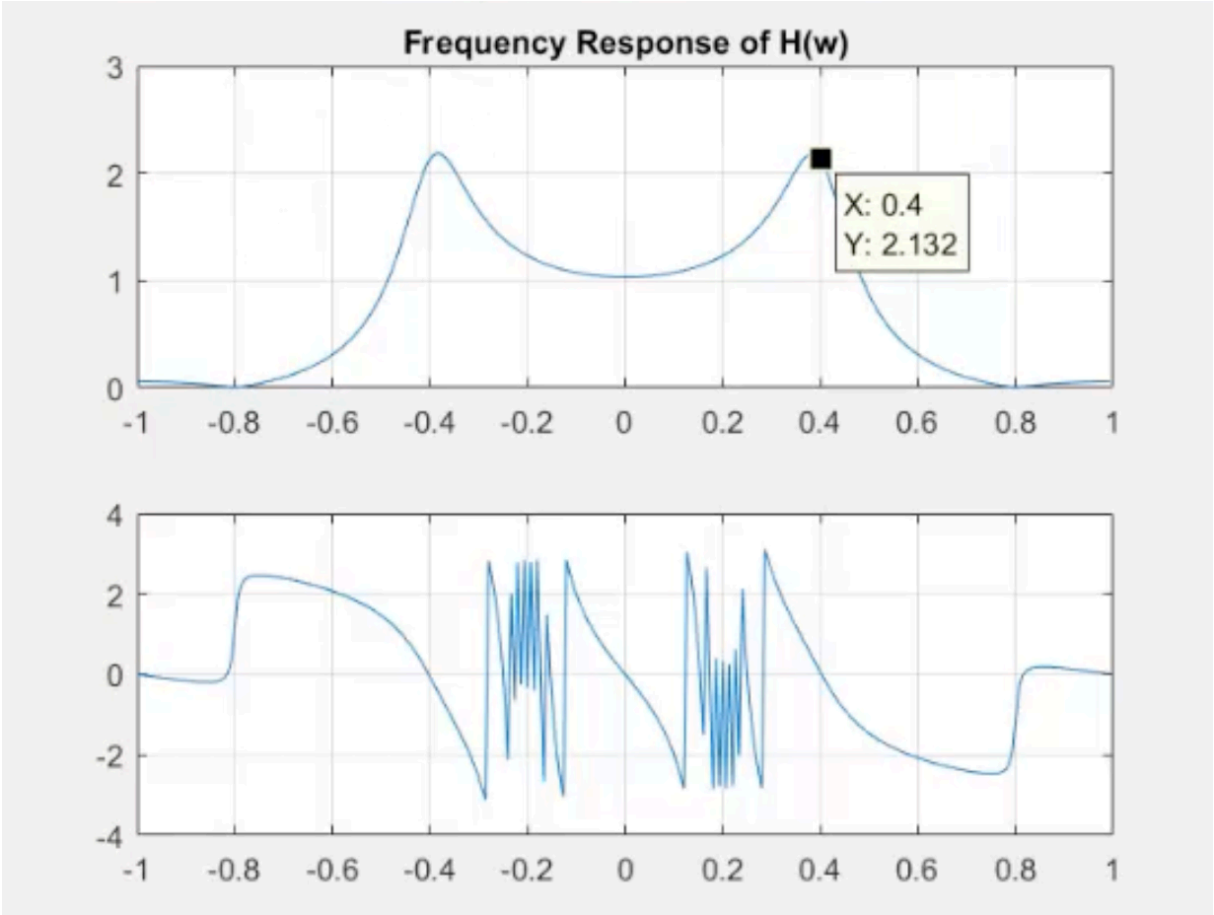
polezz = [ck, ck, conj(ck), conj(ck)]';
zerozz = 1./polezz;
K = 1;%prod(polezz);
polezz = [polezz; ps'];
zerozz = [zerozz; zs'];
Hw = Hw/3;
end
```

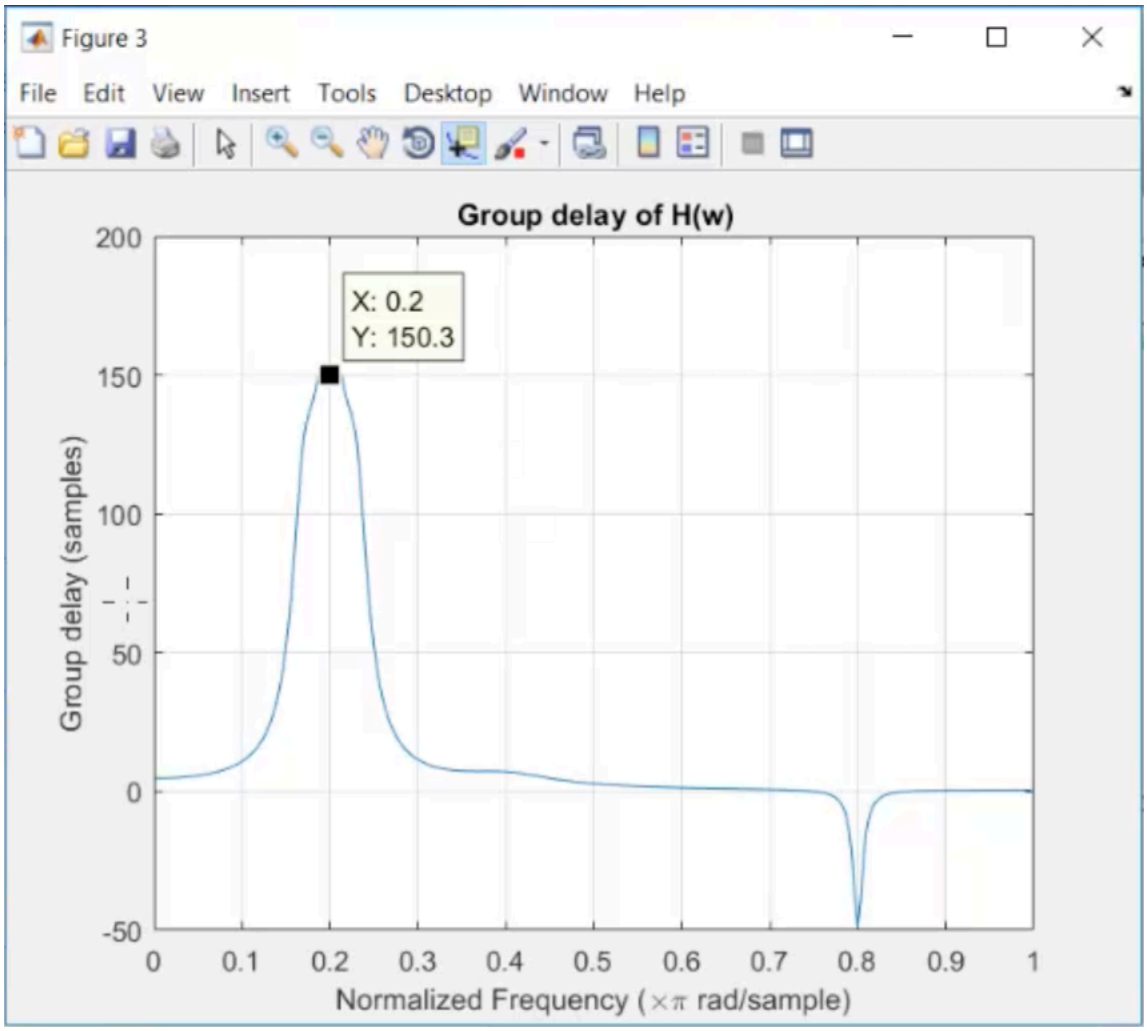
```
%% Generate original system
[Hw, p, z, K] = SystemFunction(w);
figure,
subplot 211, plot(w/pi, abs(Hw)), title('Frequency Response of H(w)'), grid on
subplot 212, plot(w/pi, (angle(Hw))), grid on

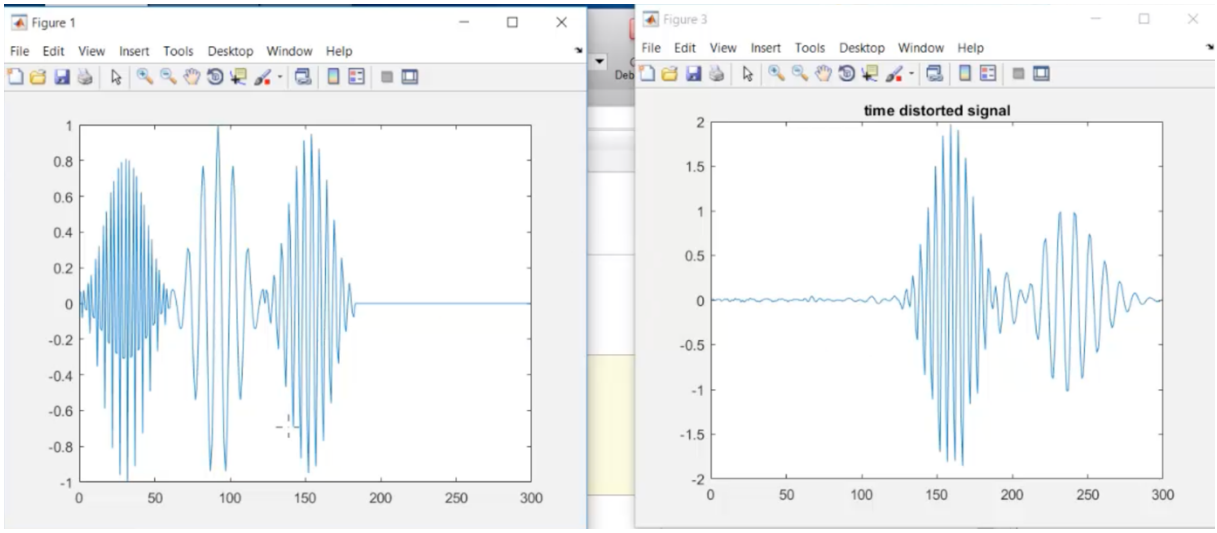
[a, b] = zp2tf(z, p, K);
a = fliplr(a); b = fliplr(b);
figure, grpdelay(b,a,N), title('Group delay of H(w)')
```


Figure 2
File Edit View Insert Tools Desktop Window Help
Frequency Response of H(w)
3
2
1
0
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1
4
2
0
-2
-4
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1


Figure 2
File Edit View Insert Tools Desktop Window Help
Frequency Response of H(w)
X: 0.2067
Y: 1.248
3
2
1
0
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1
4
2
0
-2
-4
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1


Frequency Response of H(w)
X: 0.4
Y: 2.132
3
2
1
0
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1
4
2
0
-2
-4


Figure 3
File Edit View Insert Tools Desktop Window Help
Group delay of H(w)
X: 0.2
Y: 150.3
Group delay (samples)
Normalized Frequency (×π rad/sample)


Figure 1
File Edit View Insert Tools Desktop Window Help
Figure 3
File Edit View Insert Tools Desktop Window Help
time distorted signal

3 Video 3

Linear Phase & FIR Filter Types

Nonlinear phase ⟶ group delay (non-constant) ⟶ all-pass phase compensator

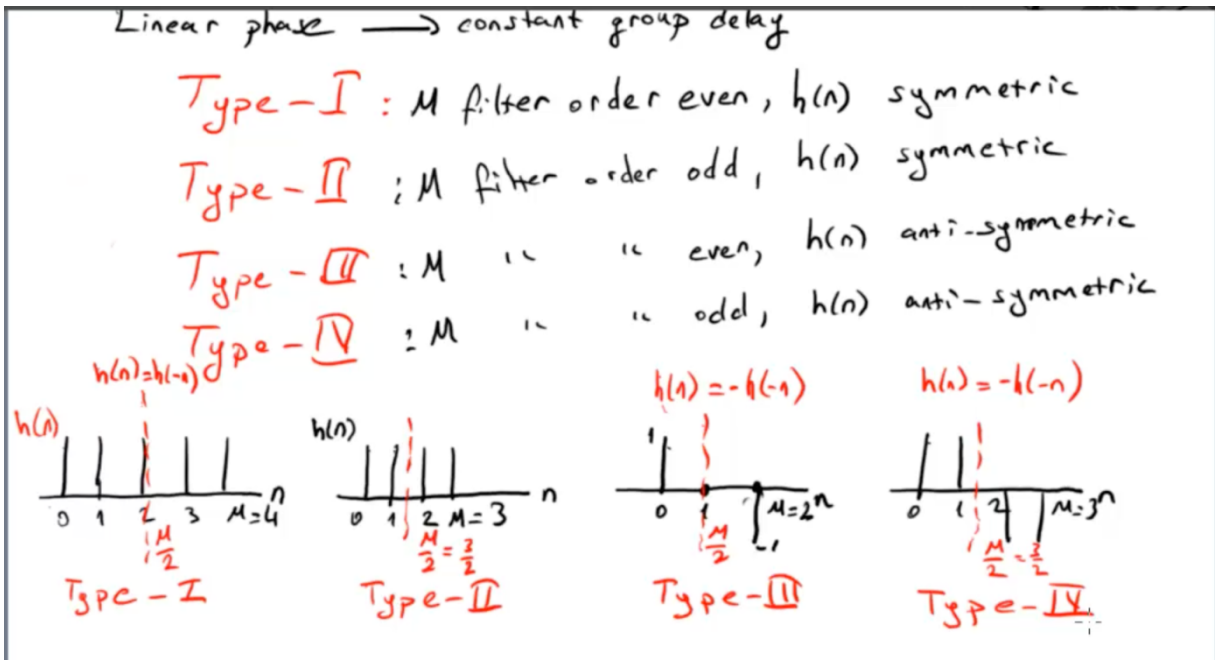
Linear phase ⟶ constant group delay

Type-I : M filter order even, h(n) symmetric

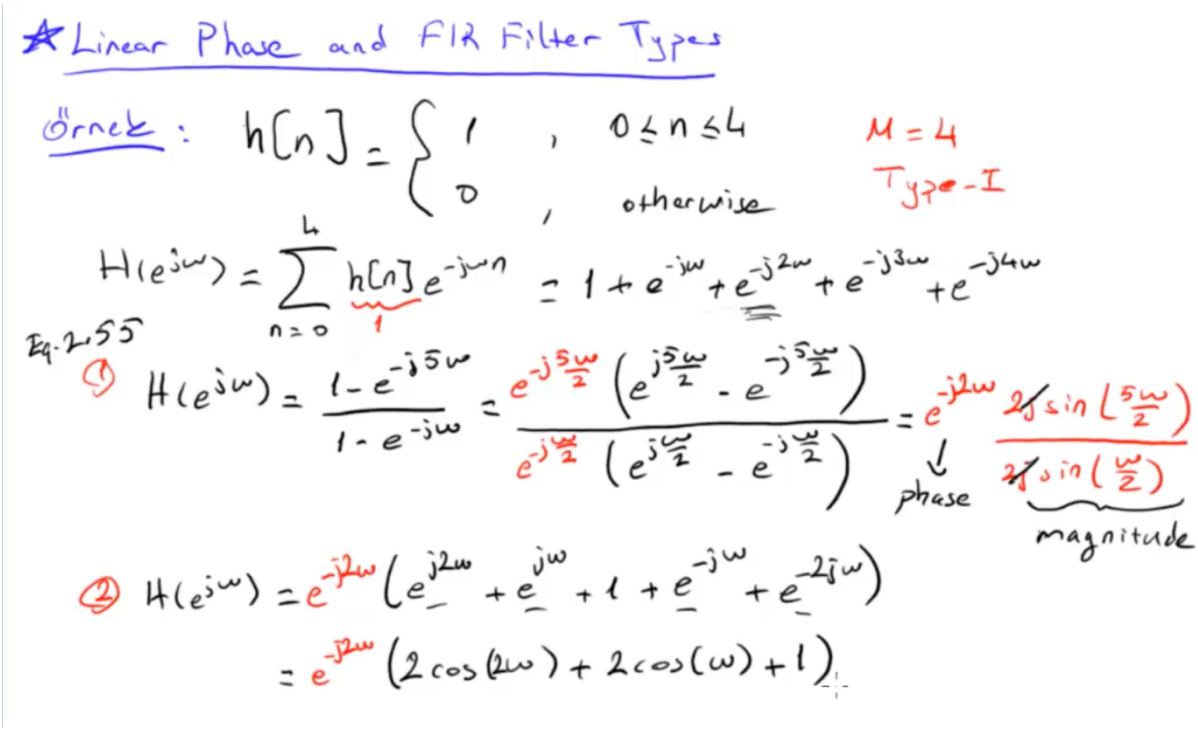
Type-II : M filter order odd, h(n) symmetric

Type-III : M " " even, h(n) anti-symmetric

Type-IV : M " " odd, h(n) anti-symmetric



4 Video 4

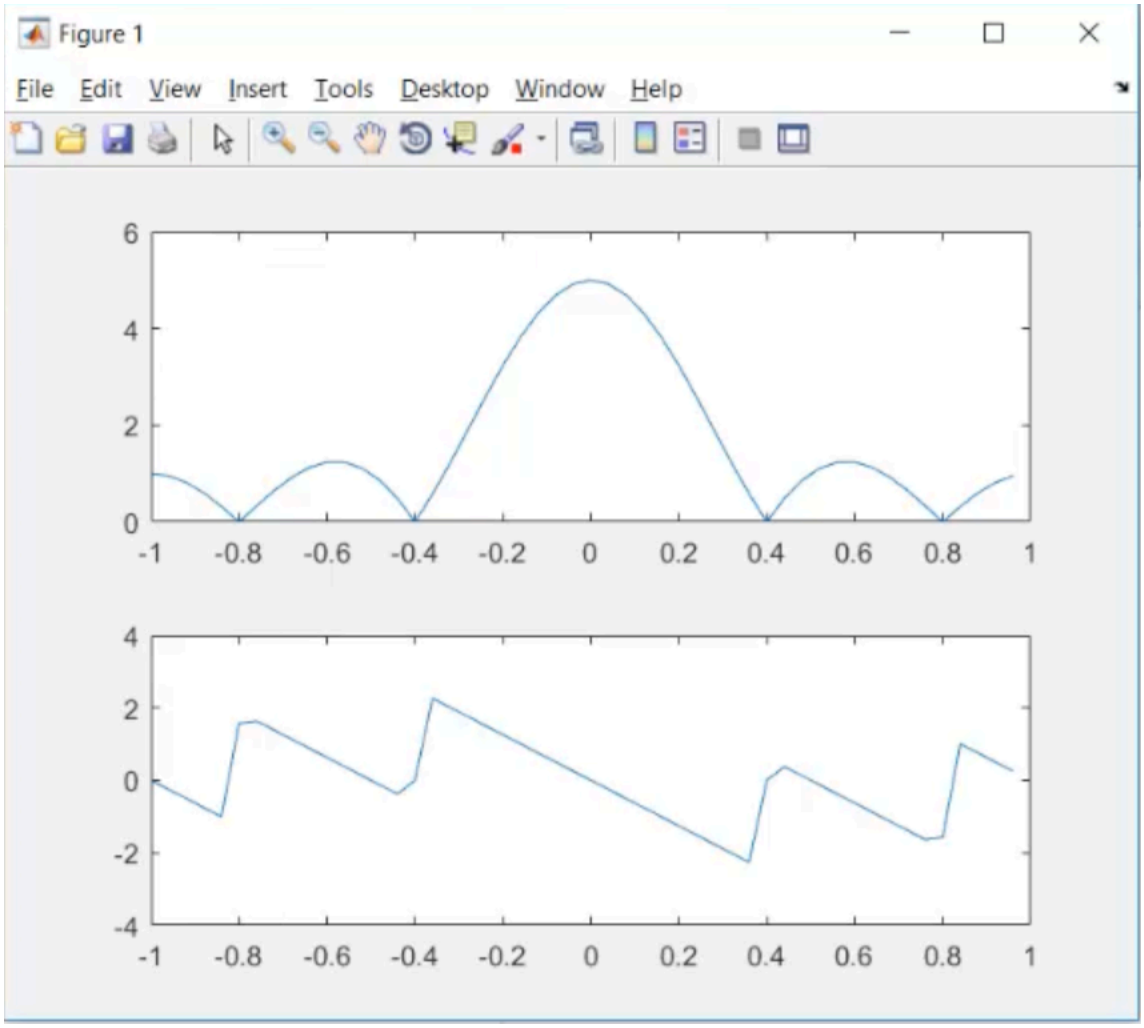


$$\tau(\omega) = -\frac{d}{d\omega}\{\angle H(e^{j\omega})\} = -\frac{d}{d\omega}\{-2\omega\} = 2 \text{ samples}$$

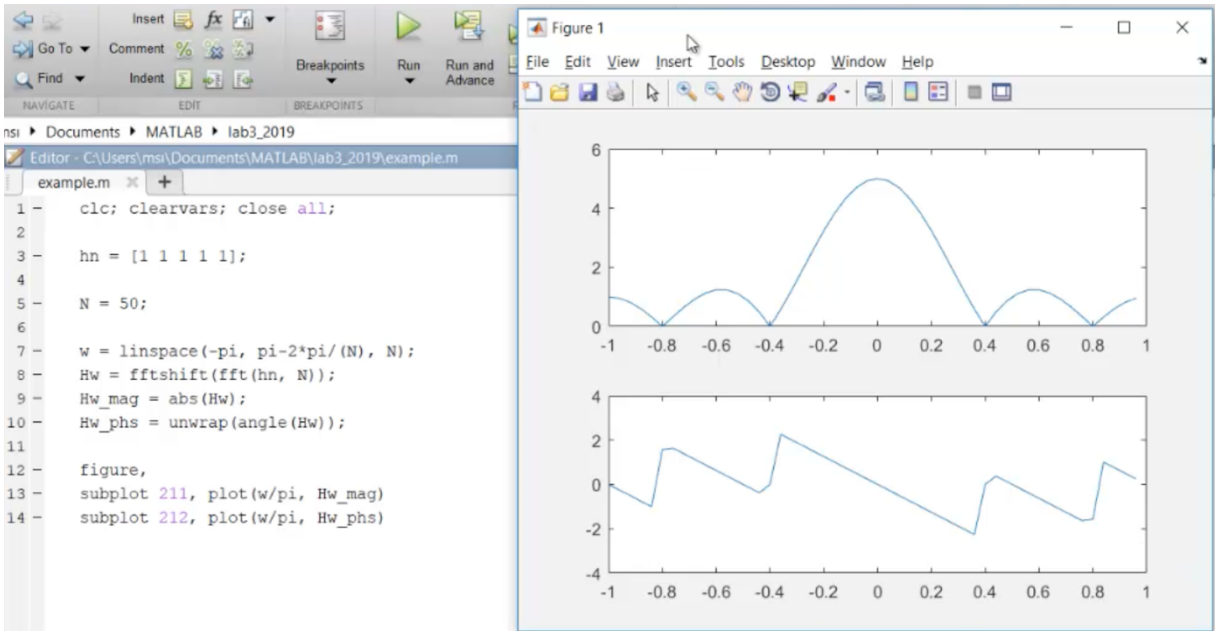
Editor - C:\Users\msi\Documents\MATLAB\lab3_2019\example.m*

example.m*

```
1 -    clc; clearvars; close all;
2
3 -    hn = [1 1 1 1 1];
4
5 -    N = 50;
6
7 -    w = linspace(-pi, pi-2*pi/(N), N);
8 -    Hw = fftshift(fft(hn, N));
9 -    Hw_mag = abs(Hw);
10 -   Hw_phs = angle(Hw);
11
12 -   figure,
13 -   subplot 211, plot(w/pi, Hw_mag)
14     subplot 212, plot(w/pi, Hw_phs)
```



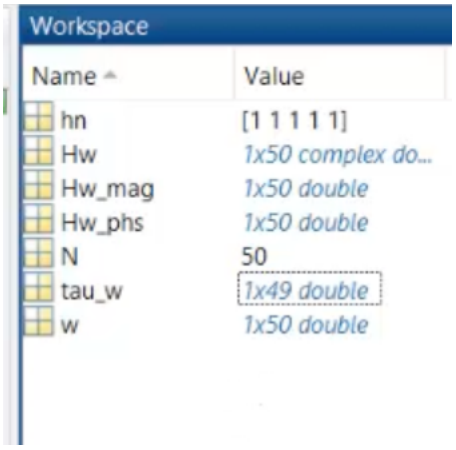
Sıkıştılmamış için unwrap



Grup gecikmesi

example.m*

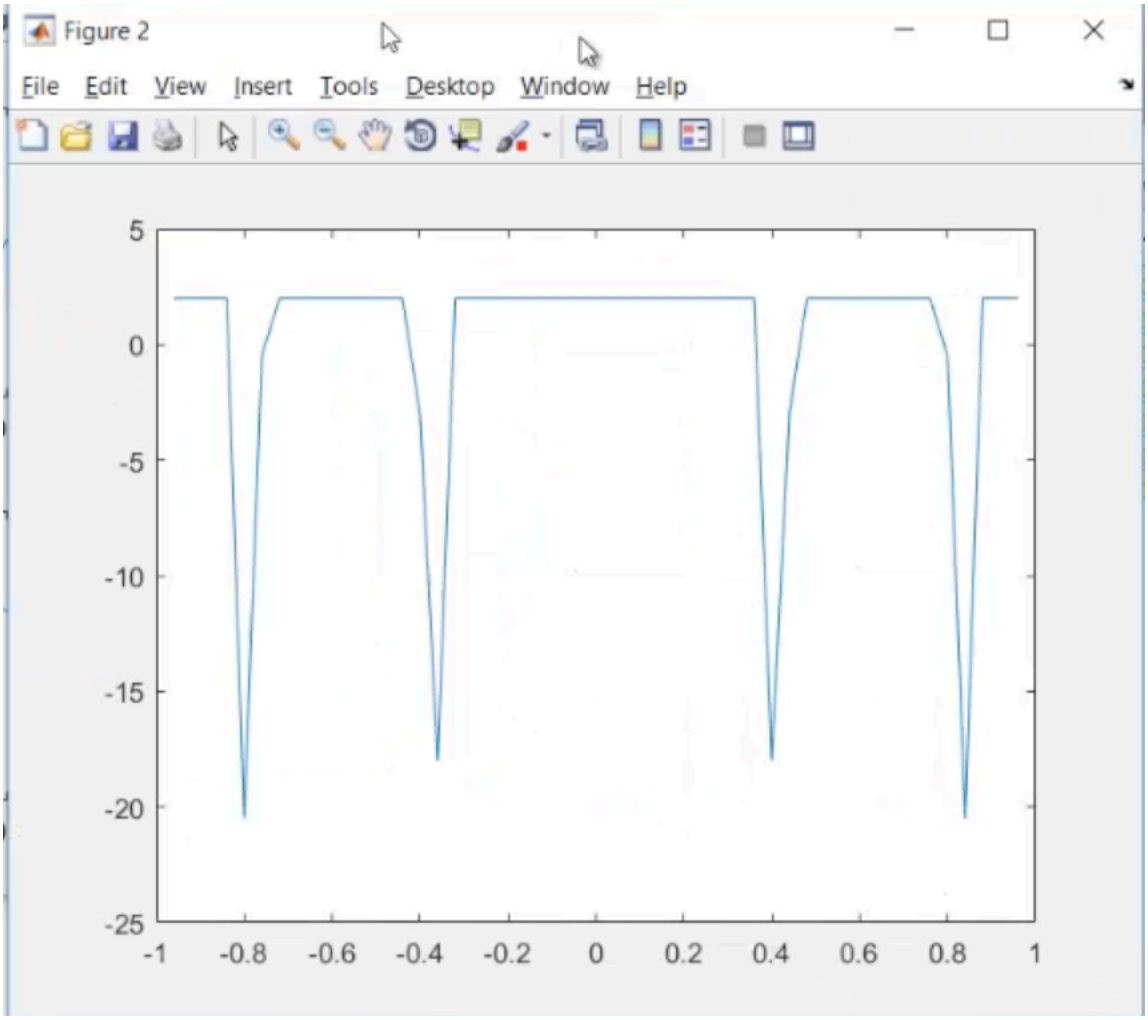
```
1 -    clc; clearvars; close all;
2
3 -    hn = [1 1 1 1 1];
4
5 -    N = 50;
6
7 -    w = linspace(-pi, pi-2*pi/(N), N);
8 -    Hw = fftshift(fft(hn, N));
9 -    Hw_mag = abs(Hw);
10 -   Hw_phs = unwrap(angle(Hw));
11
12 -   figure,
13 -   subplot 211, plot(w/pi, Hw_mag)
14 -   subplot 212, plot(w/pi, Hw_phs)
15
16 -   tau_w = -diff(Hw_phs)*(N/(2*pi));
17     figure, plot(w/pi, tau_w)
```



Sayısal türev aldığımız için tau_w bir küçüldü bu yüzden frekansta ilk örneği atıp ikinci örnekten itibaren çizdirebiliriz.

example.m*

```
1 -  clc; clearvars; close all;
2
3 -  hn = [1 1 1 1 1];
4
5 -  N = 50;
6
7 -  w = linspace(-pi, pi-2*pi/(N), N);
8 -  Hw = fftshift(fft(hn, N));
9 -  Hw_mag = abs(Hw);
10 -  Hw_phs = unwrap(angle(Hw));
11
12 -  figure,
13 -  subplot 211, plot(w/pi, Hw_mag)
14 -  subplot 212, plot(w/pi, Hw_phs)
15
16 -  tau_w = -diff(Hw_phs)*(N/(2*pi));
17 -  figure, plot(w(2:end)/pi, tau_w)
```



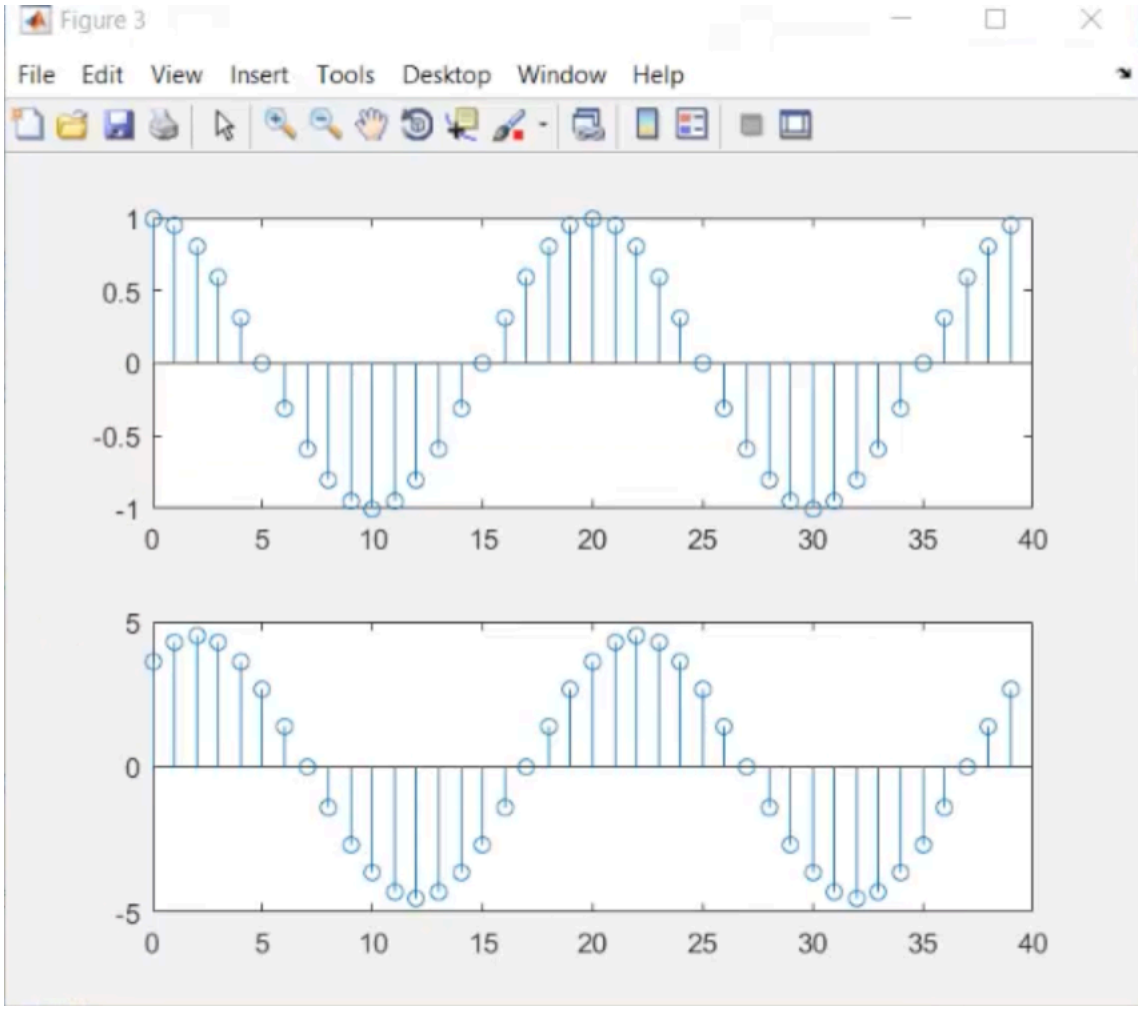
5 Video 5

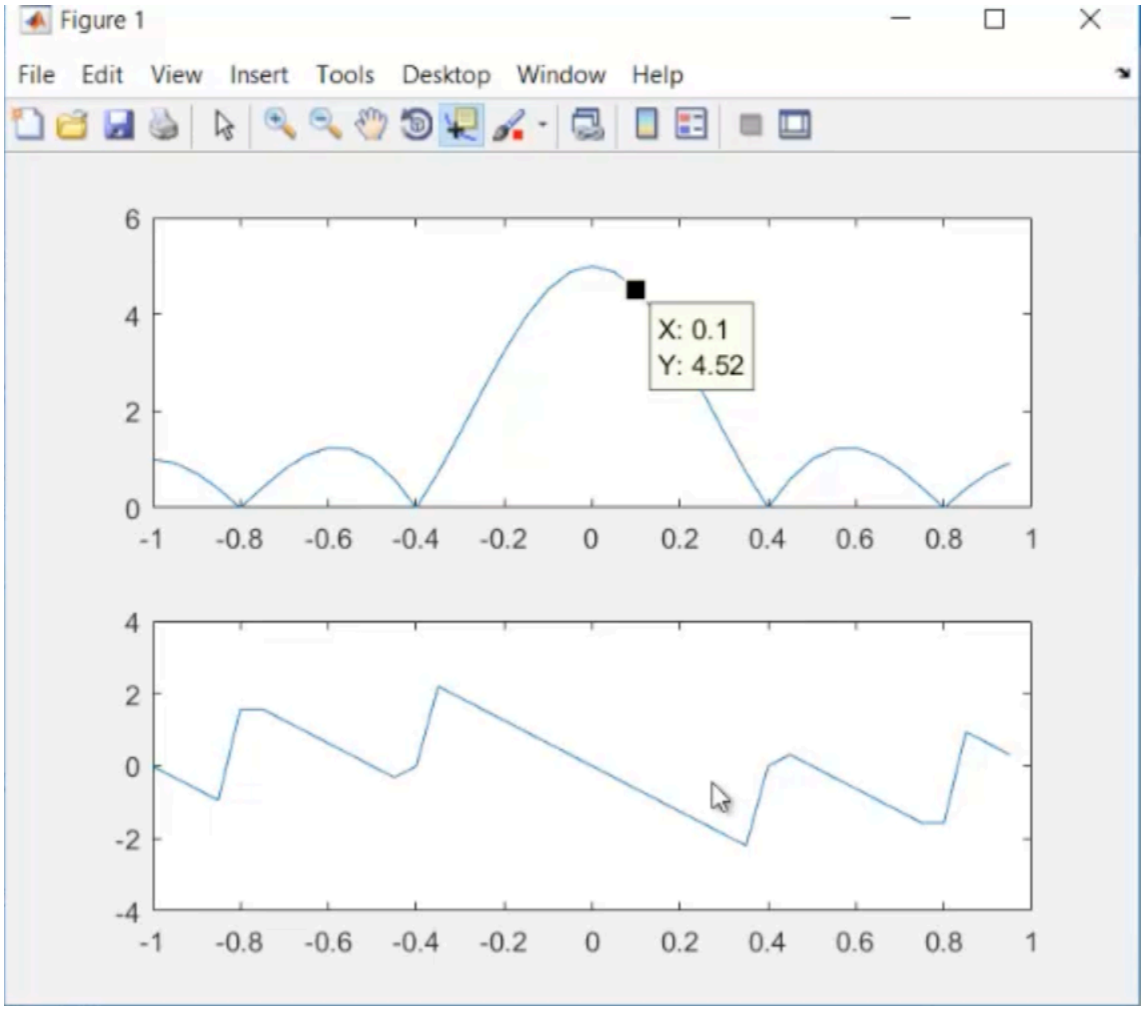
```
%% ------------------------------------------
n = 0:N-1;
w0 = .1*pi;
xn = cos(w0*n);

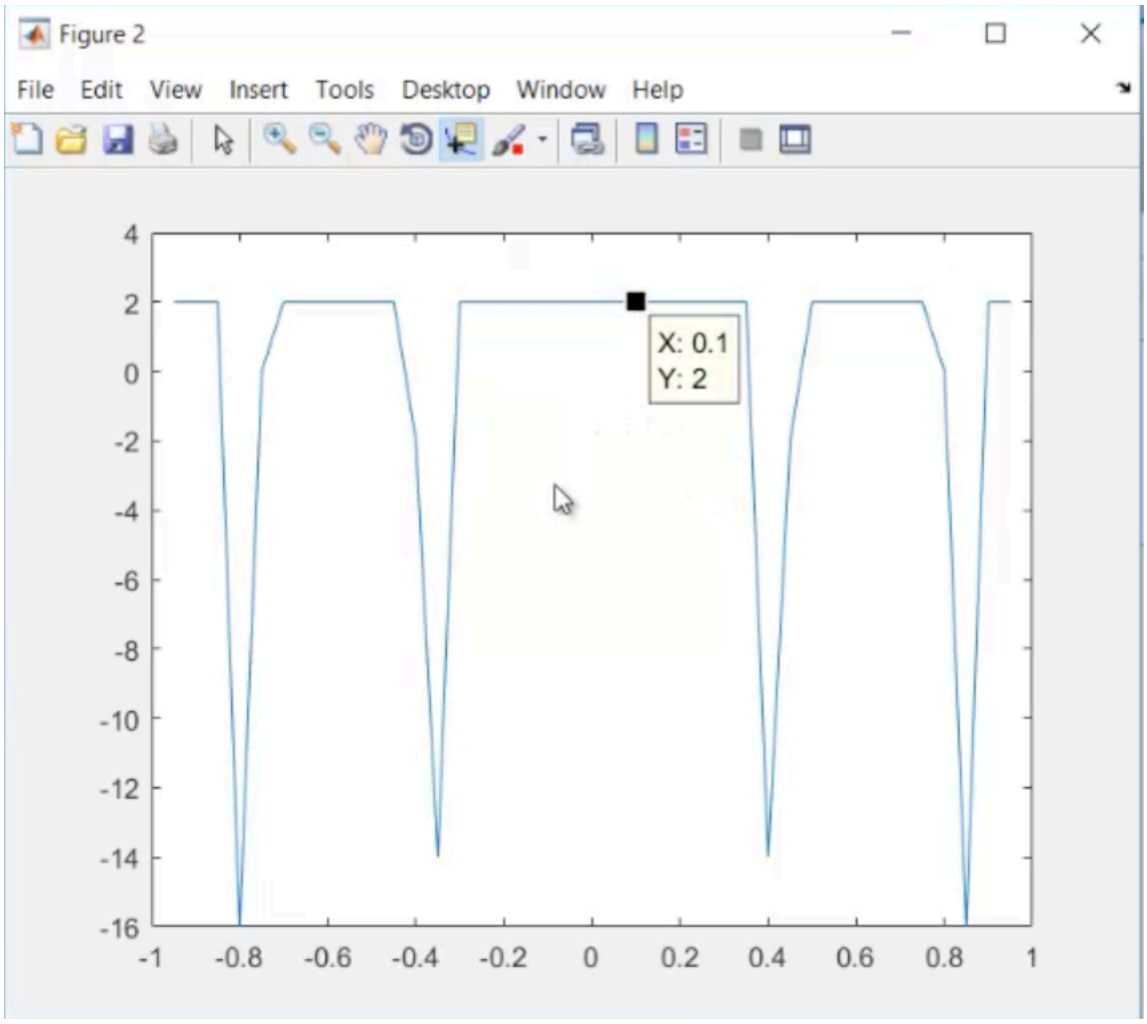
Xw = fftshift(fft(xn));

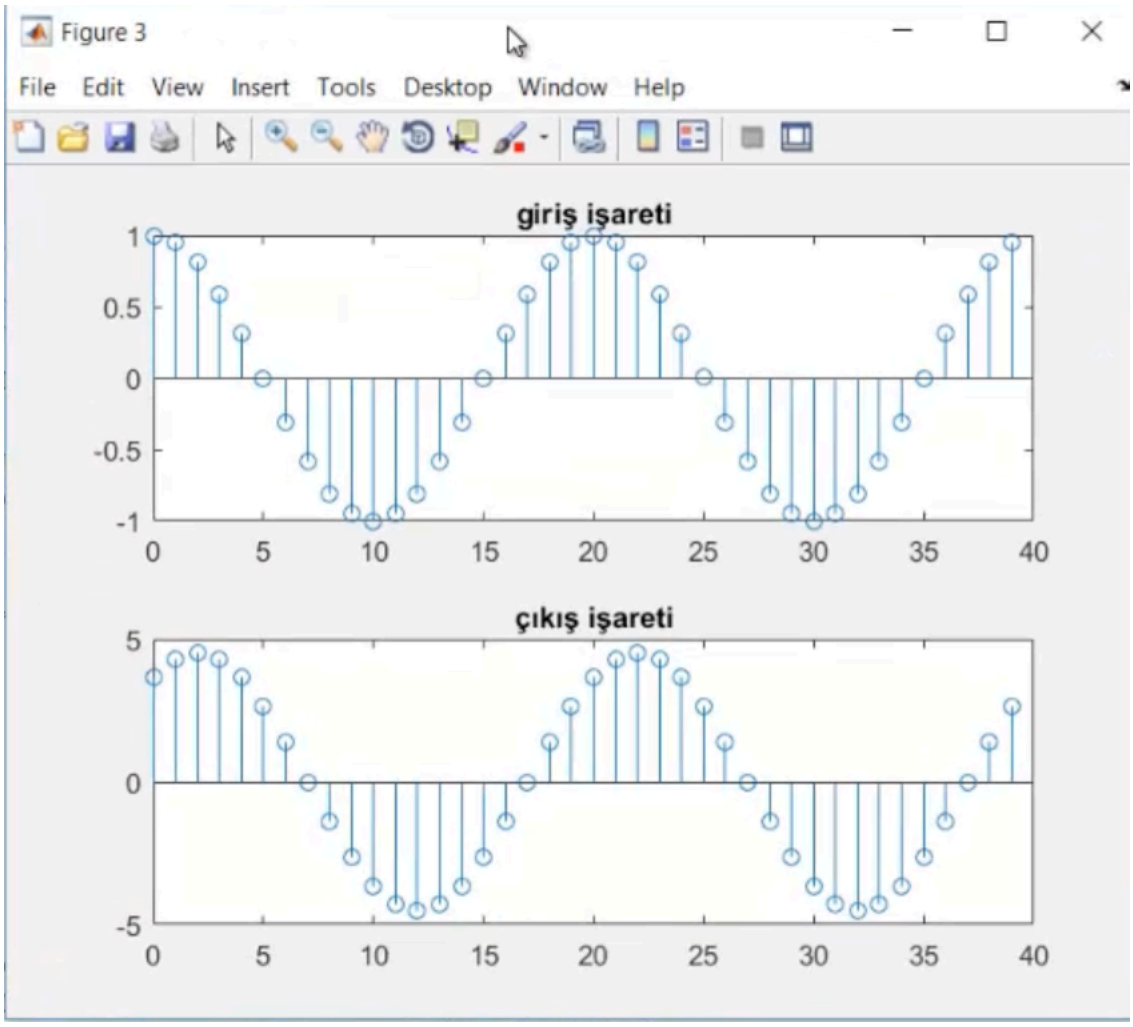
Yw = Xw.*Hw;
yn = ifft(ifftshift(Yw));

figure,
subplot 211, stem(n, xn)
subplot 212, stem(n, yn)
```




Figure 1
File Edit View Insert Tools Desktop Window Help
X: 0.1
Y: 4.52
6
4
2
0
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1
4
2
0
-2
-4
-1 -0.8 -0.6 -0.4 -0.2 0 0.2 0.4 0.6 0.8 1


Figure 2
File Edit View Insert Tools Desktop Window Help
X: 0.1
Y: 2
4
2
0
-2
-4
-6
-8
-10
-12
-14
-16
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1

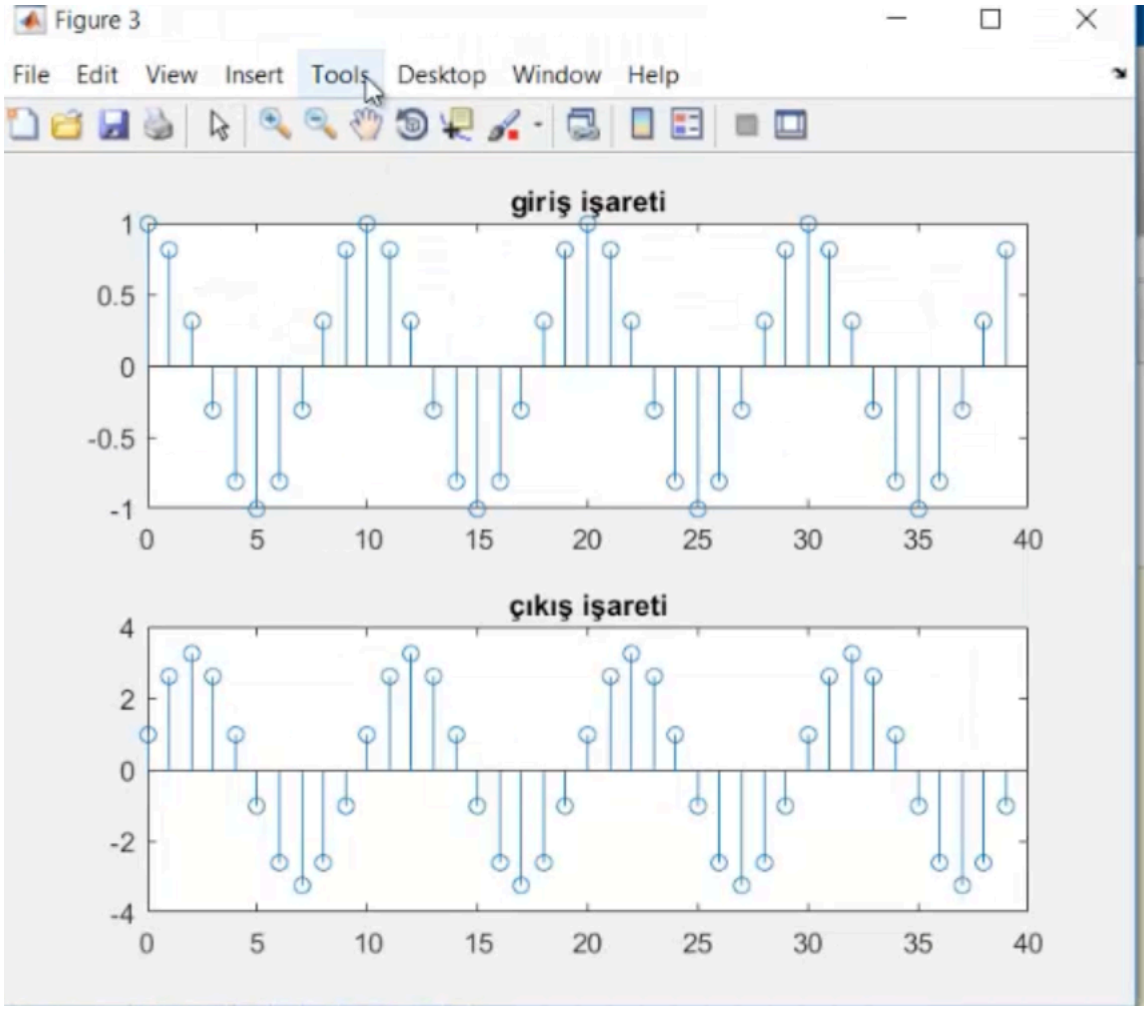



```
%% ----------------------------------------
n = 0:N-1;
w0 = .2*pi;
xn = cos(w0*n);

Xw = fftshift(fft(xn));

Yw = Xw.*Hw;
yn = ifft(ifftshift(Yw));

figure,
subplot 211, stem(n, xn), title('giriş işareti')
subplot 212, stem(n, yn), title('çıkış işareti')
```

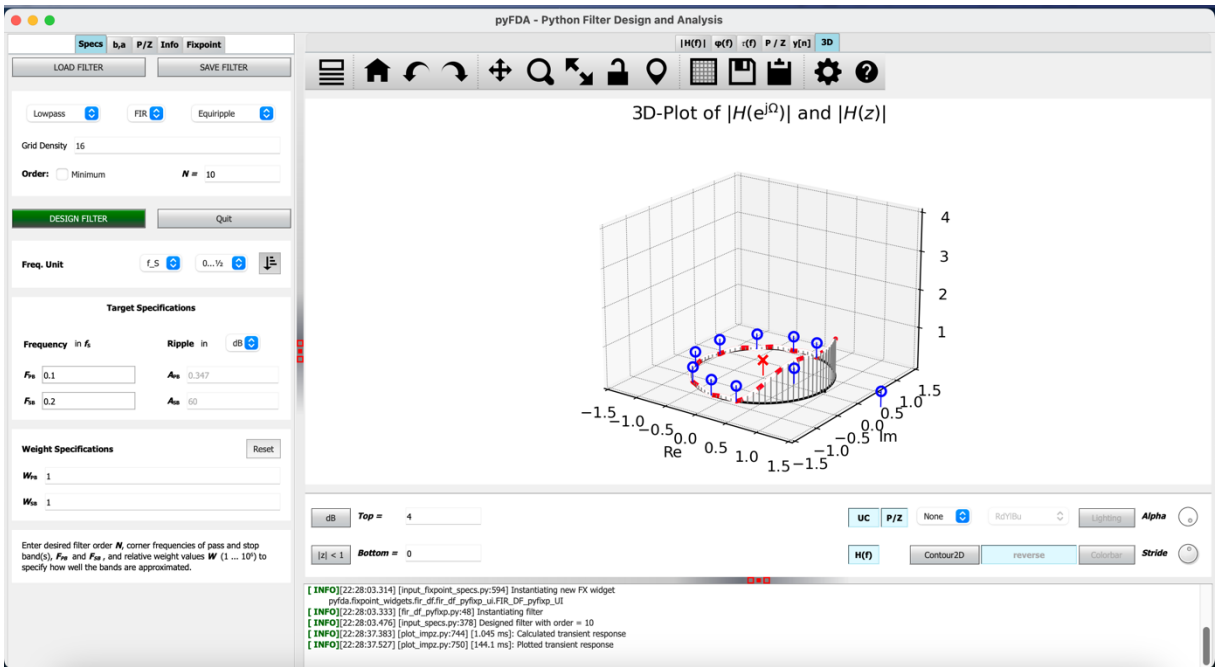


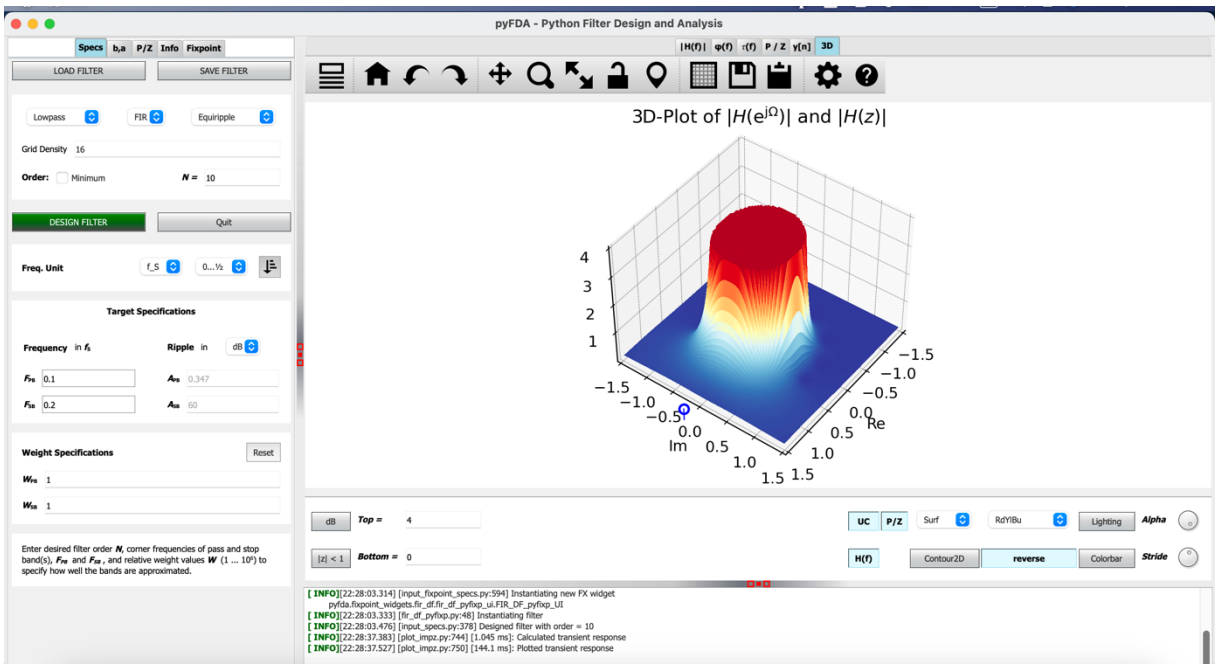
6 Video 6

```
● fuad@Fouad-MacBook-Air Digital_Signal_Processing_Python % pip install pyfda
  Collecting pyfda
    Downloading pyfda-0.8.4-py3-none-any.whl (614 kB)
       ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 614.9/614.9 kB 1.0 MB/s eta 0:00:00
  Requirement already satisfied: numpy in /opt/anaconda3/lib/python3.9/site-packages (from pyfda) (1.20.3)
  Requirement already satisfied: scipy>=1.2.0 in /opt/anaconda3/lib/python3.9/site-packages (from pyfda) (1.7.1)
  Requirement already satisfied: matplotlib>=3.1 in /opt/anaconda3/lib/python3.9/site-packages (from pyfda) (3.4.3)
  Requirement already satisfied: pyqt5 in /opt/anaconda3/lib/python3.9/site-packages (from pyfda) (5.12.3)
  Requirement already satisfied: docutils in /opt/anaconda3/lib/python3.9/site-packages (from pyfda) (0.17.1)
```

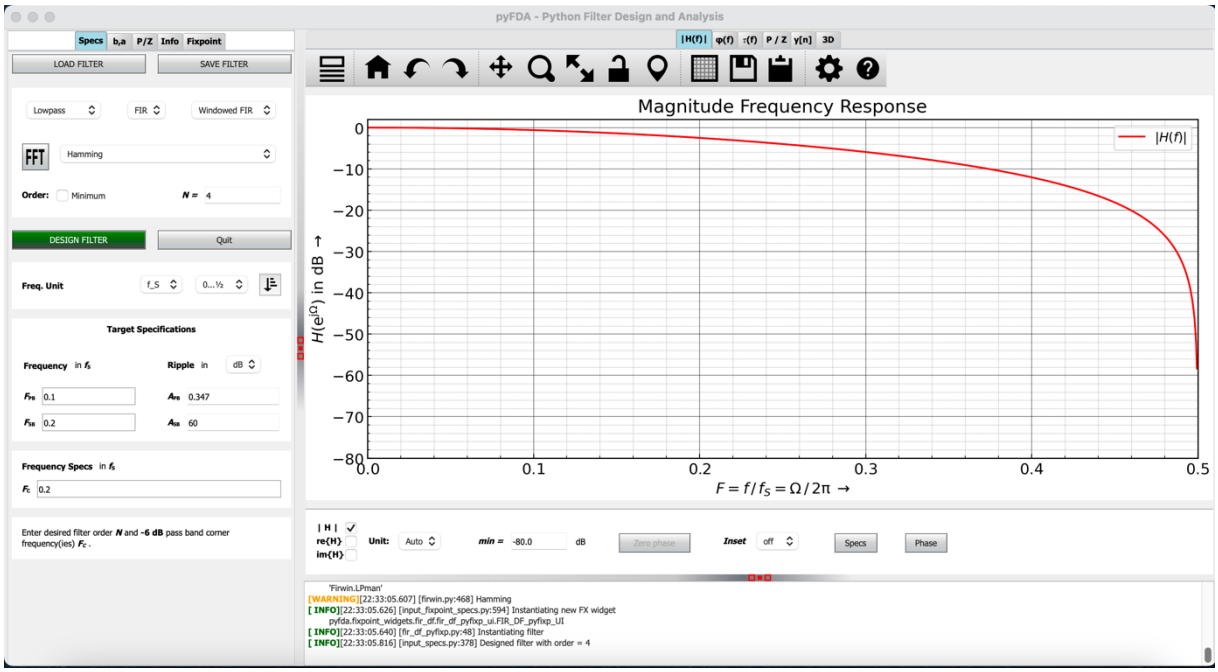
To open it

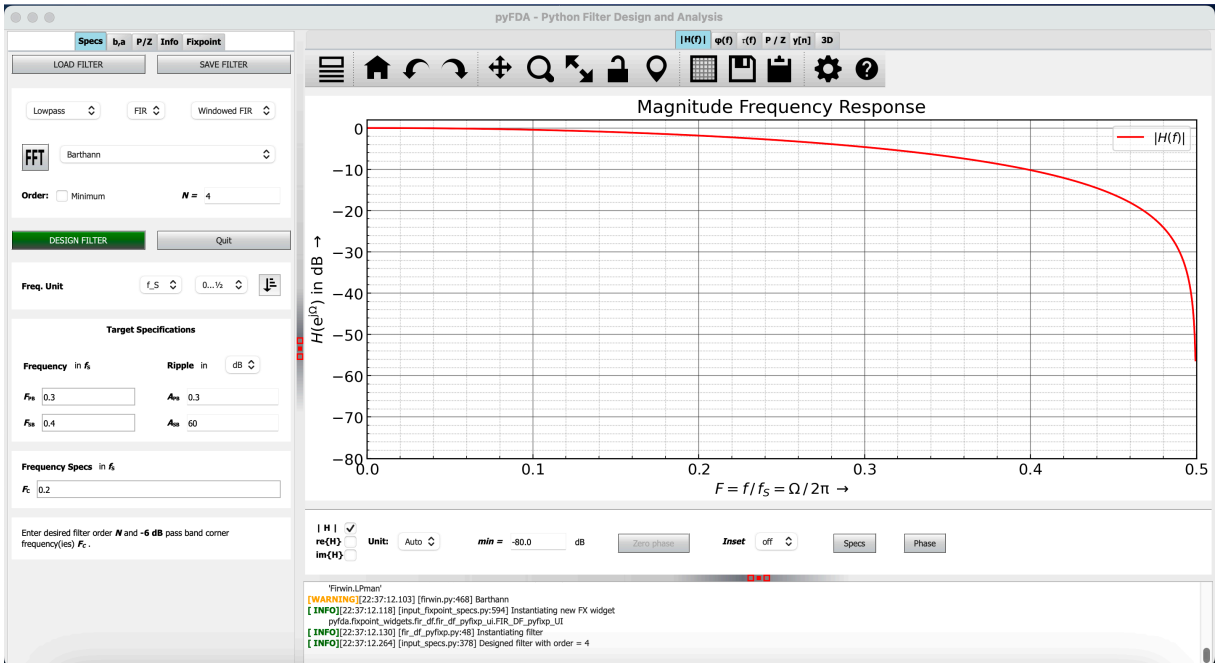
```
pyfdax
```

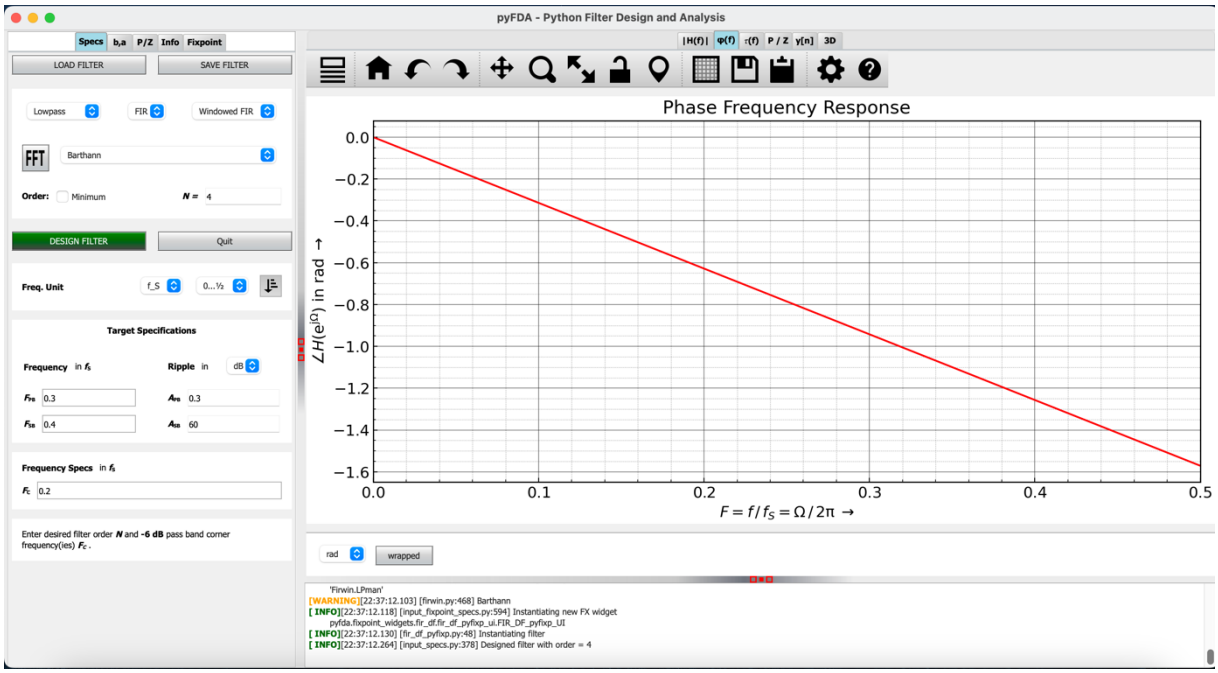

pyFDA - Python Filter Design and Analysis
Specs
b,a
P/Z
Info
Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
FIR
Equiripple
Grid Density
16
Order:
Minimum
N =
10
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_PB
0.1
A_PB
0.347
F_SB
0.2
A_SB
60
Weight Specifications
Reset
W_PB
1
W_SB
1
Enter desired filter order N, corner frequencies of pass and stop band(s), F_PB and F_SB , and relative weight values W (1 ... 10^6) to specify how well the bands are approximated.
|H(f)|
φ(f)
τ(f)
P / Z
y[n]
3D
3D-Plot of |H(e^jΩ)| and |H(z)|
Re
Im
dB
Top =
4
|z| < 1
Bottom =
0
UC
P/Z
None
RdYlBu
Lighting
Alpha
H(f)
Contour2D
reverse
Colorbar
Stride
[INFO][22:28:03.314] [input_fixpoint_specs.py:594] Instantiating new FX widget
pyfda.fixpoint_widgets.fir_df.fir_df_pyfixp_ui.FIR_DF_pyfixp_UI
[INFO][22:28:03.333] [fir_df_pyfixp.py:48] Instantiating filter
[INFO][22:28:03.476] [input_specs.py:378] Designed filter with order = 10
[INFO][22:28:37.383] [plot_impz.py:744] [1.045 ms]: Calculated transient response
[INFO][22:28:37.527] [plot_impz.py:750] [144.1 ms]: Plotted transient response

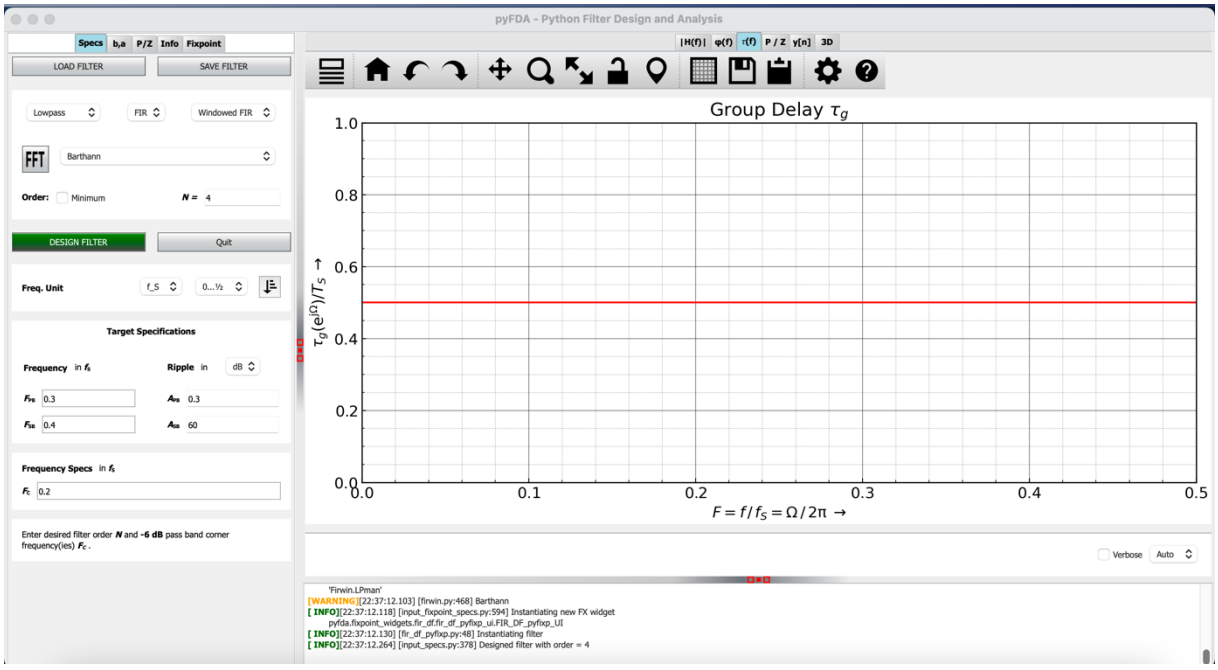

pyFDA - Python Filter Design and Analysis
Specs
b,a
P/Z
Info
Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
FIR
Equiripple
Grid Density
16
Order:
Minimum
N =
10
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_PB
0.1
A_PB
0.347
F_SB
0.2
A_SB
60
Weight Specifications
Reset
W_PB
1
W_SB
1
Enter desired filter order N, corner frequencies of pass and stop band(s), F_PB and F_SB , and relative weight values W (1 ... 10^6) to specify how well the bands are approximated.
|H(f)|
φ(f)
τ(f)
P / Z
y[n]
3D
3D-Plot of |H(e^jΩ)| and |H(z)|
Re
Im
dB
Top =
4
|z| < 1
Bottom =
0
UC
P/Z
Surf
RdYlBu
Lighting
Alpha
H(f)
Contour2D
reverse
Colorbar
Stride
[INFO][22:28:03.314] [input_fixpoint_specs.py:594] Instantiating new FX widget
pyfda.fixpoint_widgets.fir_df.fir_df_pyfixp_ui.FIR_DF_pyfixp_UI
[INFO][22:28:03.333] [fir_df_pyfixp.py:48] Instantiating filter
[INFO][22:28:03.476] [input_specs.py:378] Designed filter with order = 10
[INFO][22:28:37.383] [plot_impz.py:744] [1.045 ms]: Calculated transient response
[INFO][22:28:37.527] [plot_impz.py:750] [144.1 ms]: Plotted transient response

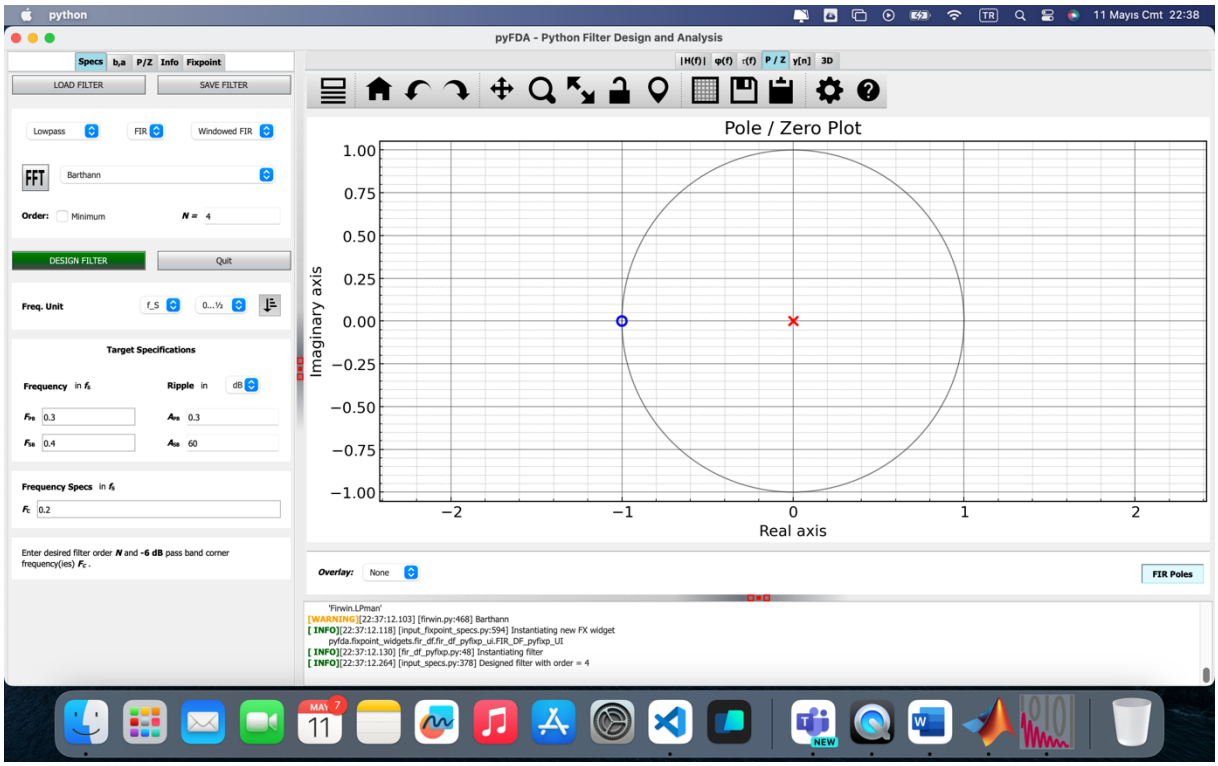
0 ve pi arasında

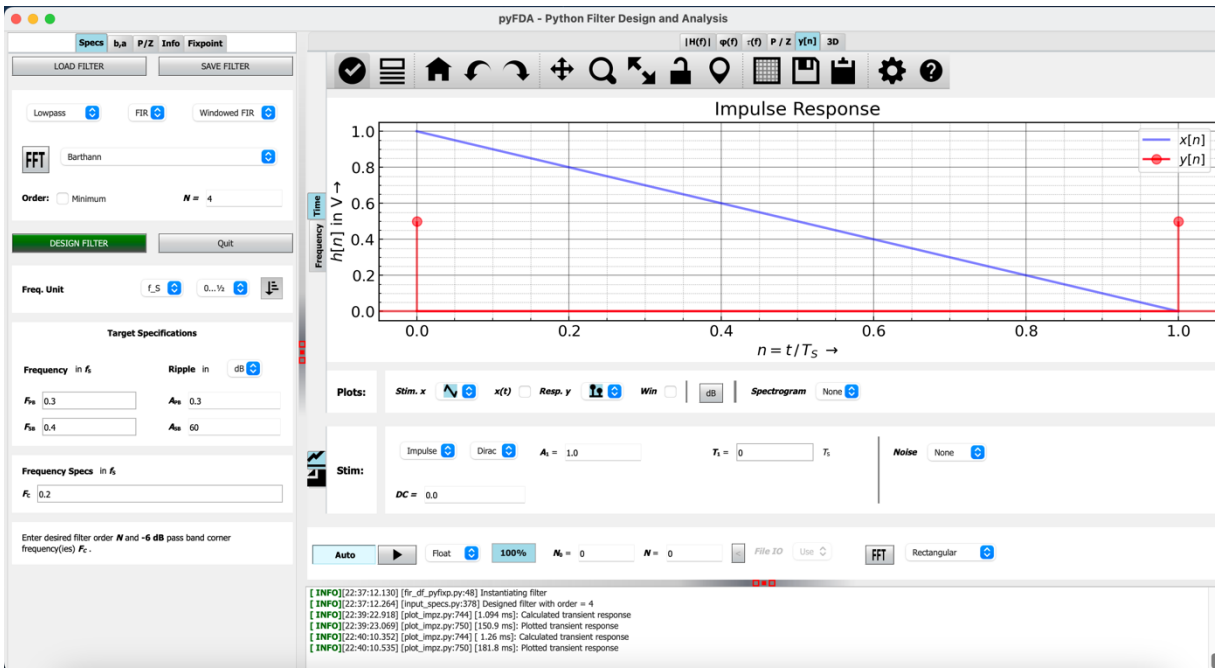




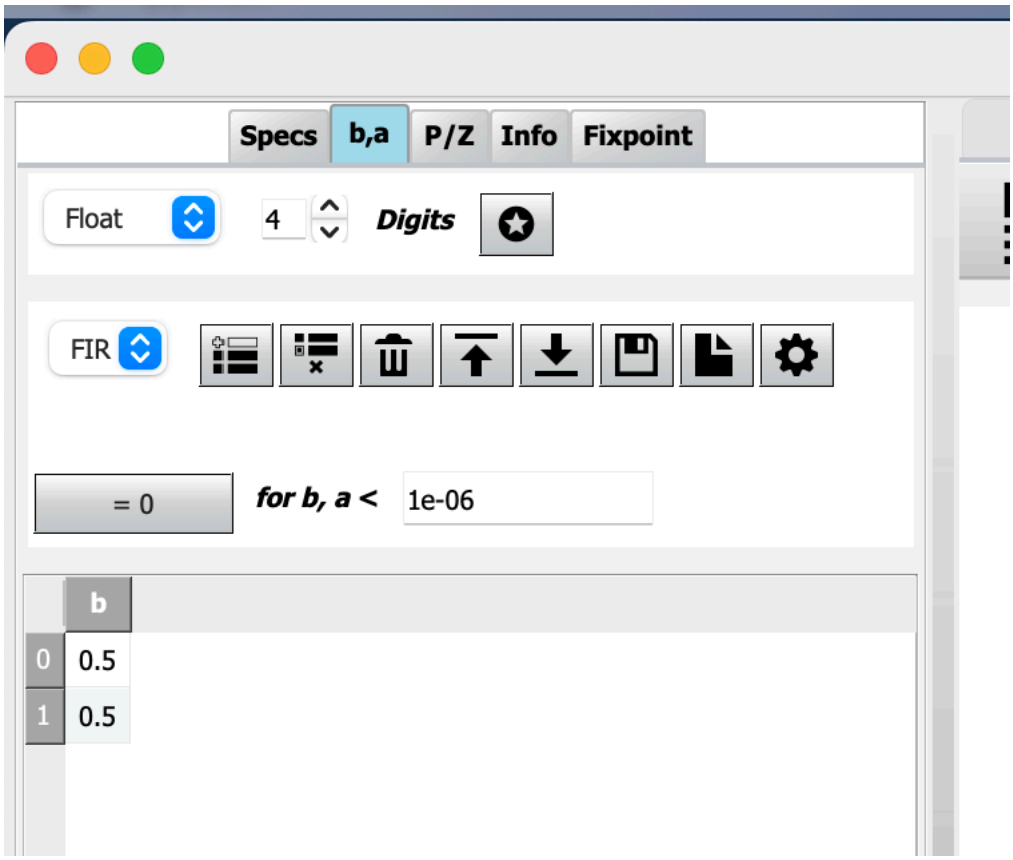

pyFDA - Python Filter Design and Analysis
Specs
b,a
P/Z
Info
Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
FIR
Windowed FIR
FFT
Barthann
Order:
Minimum
N =
4
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_SB
0.3
A_PB
0.3
F_SB
0.4
A_SB
60
Frequency Specs in f_S
F_C
0.2
Enter desired filter order N and -6 dB pass band corner frequency(ies) F_C .
|H(f)|
φ(f)
τ(f)
P / Z
y[n]
3D
Phase Frequency Response
∠H(e^jΩ) in rad →
F = f/f_S = Ω/2π →
rad
wrapped
'Firwin.LPman'
[WARNING][22:37:12.103] [firwin.py:468] Barthann
[INFO][22:37:12.118] [input_fixpoint_specs.py:594] Instantiating new FX widget
pyfda.fixpoint_widgets.fir_df.fir_df_pyfixp_ui.FIR_DF_pyfixp_UI
[INFO][22:37:12.130] [fir_df_pyfixp.py:48] Instantiating filter
[INFO][22:37:12.264] [input_specs.py:378] Designed filter with order = 4


pyFDA - Python Filter Design and Analysis
Specs
b,a
P/Z
Info
Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
FIR
Windowed FIR
FFT
Barthann
Order:
Minimum
N =
4
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_SB
0.3
A_PB
0.3
F_SB
0.4
A_SB
60
Frequency Specs in f_S
F_C
0.2
Enter desired filter order N and -6 dB pass band corner frequency(ies) F_C .
|H(f)|
φ(f)
τ(f)
P / Z
y[n]
3D
Group Delay τ_g
τ_g(e^jΩ)/T_S →
F = f/f_S = Ω/2π →
Verbose
Auto
'Firwin.LPman'
[WARNING][22:37:12.103] [firwin.py:468] Barthann
[INFO][22:37:12.118] [input_fixpoint_specs.py:594] Instantiating new FX widget
pyfda.fixpoint_widgets.fir_df.fir_df_pyfixp_ui.FIR_DF_pyfixp_UI
[INFO][22:37:12.130] [fir_df_pyfixp.py:48] Instantiating filter
[INFO][22:37:12.264] [input_specs.py:378] Designed filter with order = 4

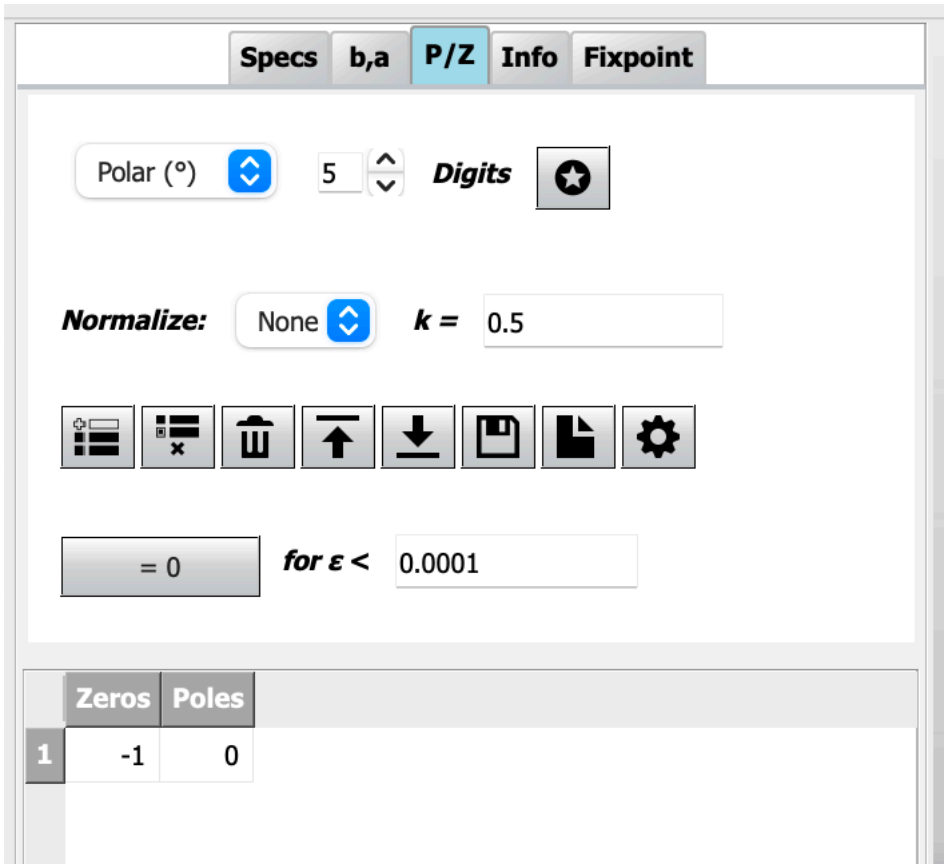

python
11 Mayıs Cmt 22:38
pyFDA - Python Filter Design and Analysis
Specs b,a P/Z Info Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
FIR
Windowed FIR
FFT
Barthann
Order: Minimum
N = 4
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_PB 0.3
A_PB 0.3
F_SB 0.4
A_SB 60
Frequency Specs in f_S
F_C 0.2
Enter desired filter order N and -6 dB pass band corner frequency(ies) F_C .
|H(f)| φ(f) τ(f) P / Z y[n] 3D
Pole / Zero Plot
Imaginary axis
Real axis
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−2
−1
0
1
2
Overlay: None
FIR Poles
'Firwin.LPman'
[WARNING][22:37:12.103] [firwin.py:468] Barthann
[INFO][22:37:12.118] [input_fixpoint_specs.py:594] Instantiating new FX widget
pyfda.fixpoint_widgets.fir_df.fir_df_pyfixp_ui.FIR_DF_pyfixp_UI
[INFO][22:37:12.130] [fir_df_pyfixp.py:48] Instantiating filter
[INFO][22:37:12.264] [input_specs.py:378] Designed filter with order = 4

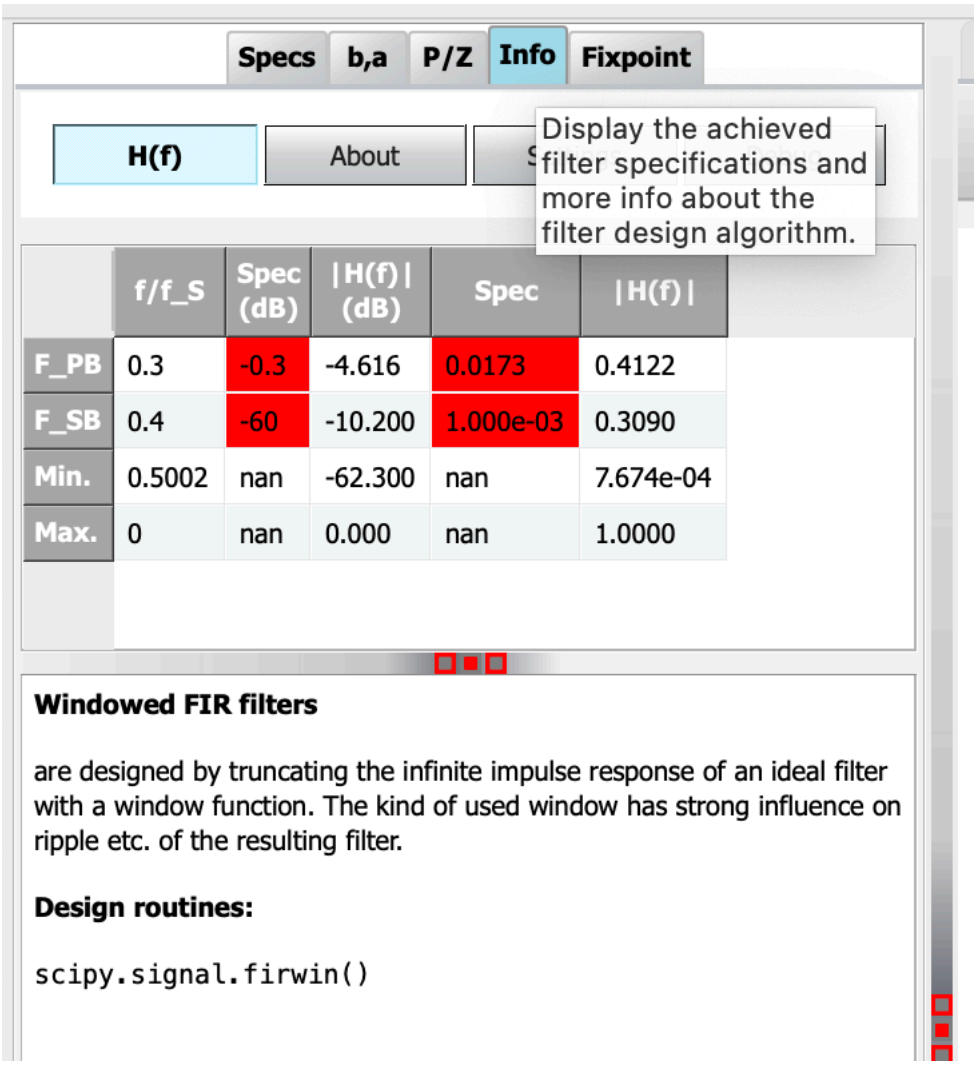

pyFDA - Python Filter Design and Analysis
Specs b,a P/Z Info Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
FIR
Windowed FIR
FFT
Barthann
Order: Minimum
N = 4
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_PB 0.3
A_PB 0.3
F_SB 0.4
A_SB 60
Frequency Specs in f_S
F_C 0.2
Enter desired filter order N and -6 dB pass band corner frequency(ies) F_C .
|H(f)| φ(f) τ(f) P / Z y[n] 3D
Impulse Response
x[n]
y[n]
h[n] in V →
n = t / T_S →
Time
Frequency
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Plots:
Stim. x
x(t)
Resp. y
Win
dB
Spectrogram
None
Stim:
Impulse
Dirac
A_1 = 1.0
T_1 = 0
T_S
Noise
None
DC = 0.0
Auto
Float
100%
N_B = 0
N = 0
File IO
Use
FFT
Rectangular
[INFO][22:37:12.130] [fir_df_pyfixp.py:48] Instantiating filter
[INFO][22:37:12.264] [input_specs.py:378] Designed filter with order = 4
[INFO][22:39:22.918] [plot_impz.py:744] [1.094 ms]: Calculated transient response
[INFO][22:39:23.069] [plot_impz.py:750] [150.9 ms]: Plotted transient response
[INFO][22:40:10.352] [plot_impz.py:744] [1.26 ms]: Calculated transient response
[INFO][22:40:10.535] [plot_impz.py:750] [181.8 ms]: Plotted transient response

Filtrenin katsayıları



Kutup sıfır




Specs
b,a
P/Z
Info
Fixpoint
H(f)
About
Display the achieved filter specifications and more info about the filter design algorithm.
f/f_S
Spec (dB)
|H(f)| (dB)
Spec
|H(f)|
F_PB
0.3
-0.3
-4.616
0.0173
0.4122
F_SB
0.4
-60
-10.200
1.000e-03
0.3090
Min.
0.5002
nan
-62.300
nan
7.674e-04
Max.
0
nan
0.000
nan
1.0000
Windowed FIR filters
are designed by truncating the infinite impulse response of an ideal filter with a window function. The kind of used window has strong influence on ripple etc. of the resulting filter.
Design routines:
scipy.signal.firwin()

Specs | b,a | P/Z | Info | Fixpoint

FIR_DF (pyfixp) **Direct-Form (DF) FIR Filter**

Standard FIR topology, suitable for most use cases.

Sim. FX

Input Quantizer Q_X: 0 . 15

Wrap Round

Coeff. Quantization $b_{I.F}$: 0 . 15

Wrap Round

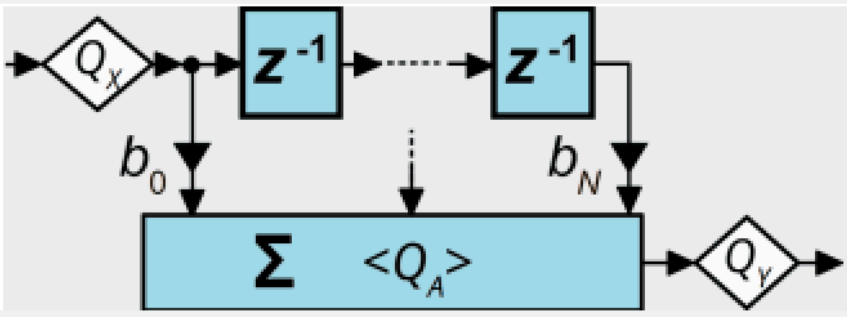
Max_{10} = 1 LSB_{10} = 3.052e-05

Accu Format Q_A: A 0 . 30

Wrap Round

Output Quantizer Q_Y: 0 . 15

Wrap Round

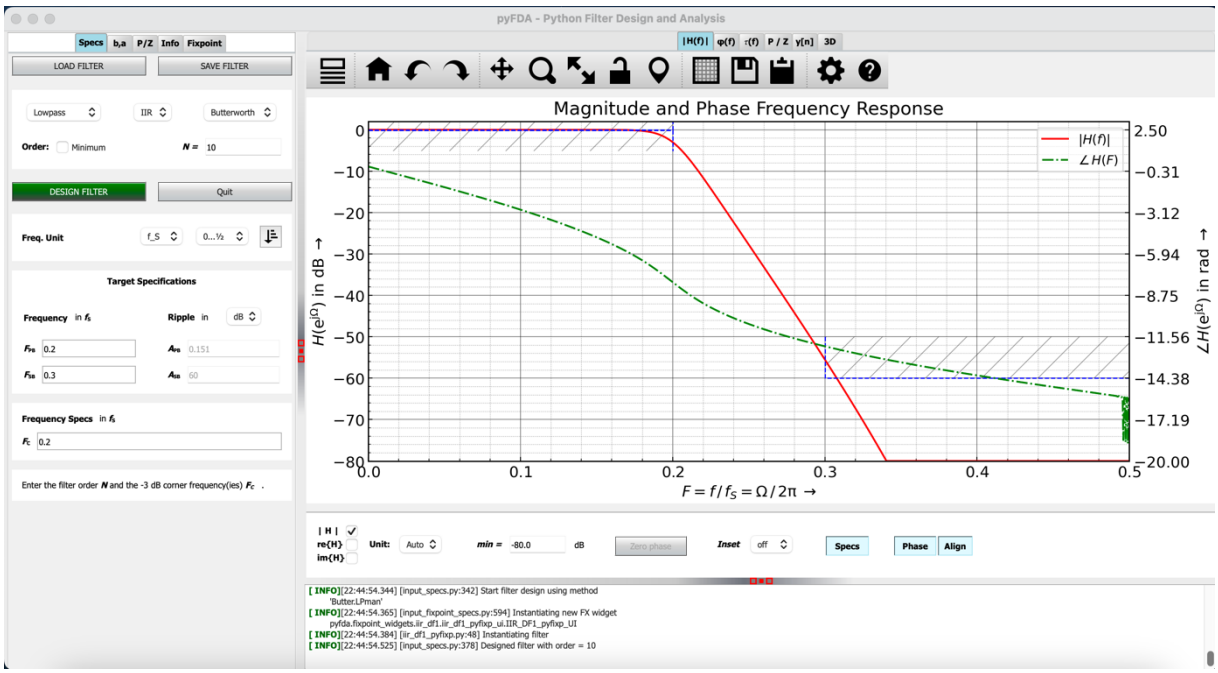


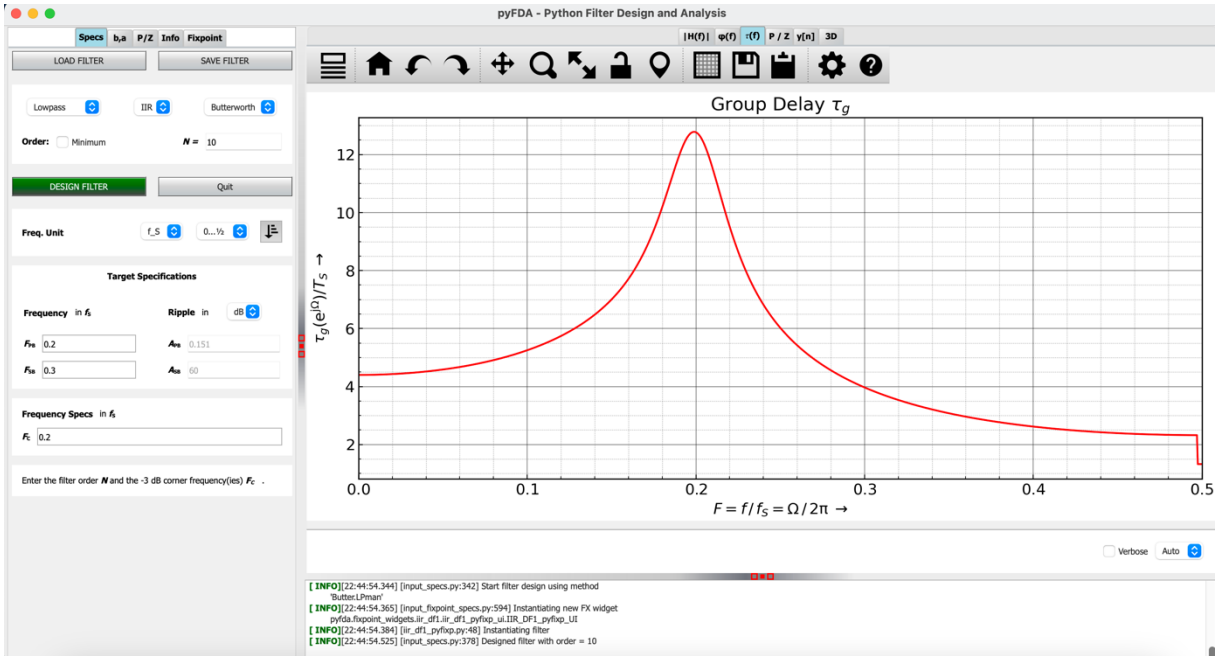
dB

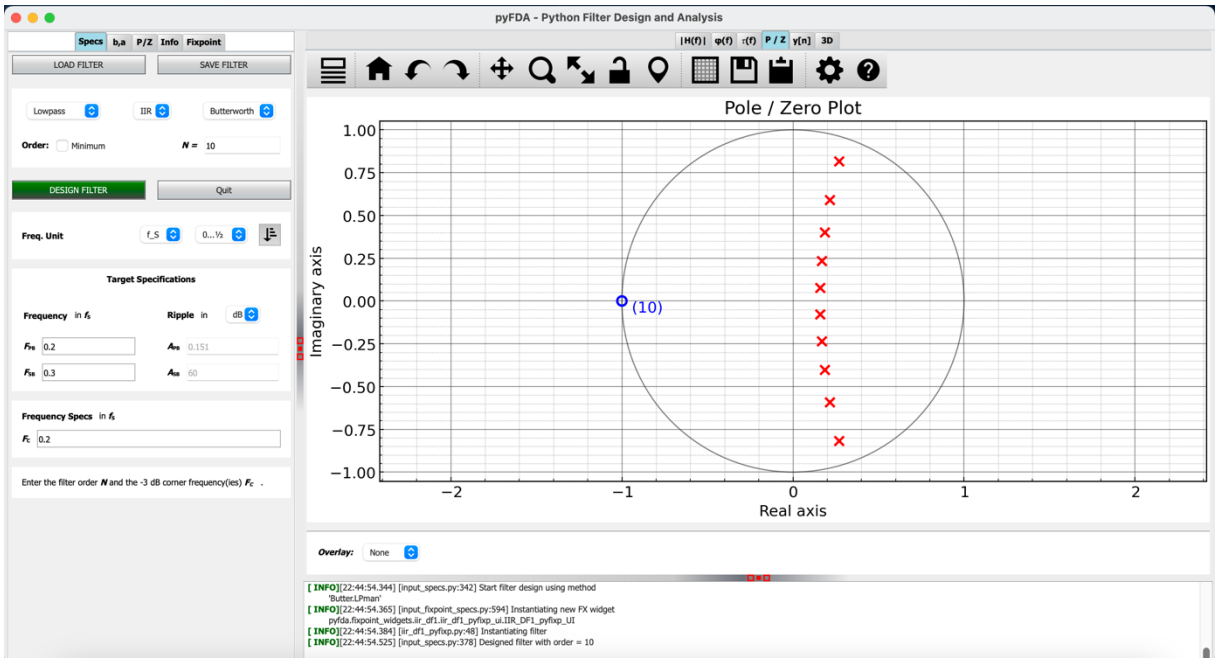
|z| <

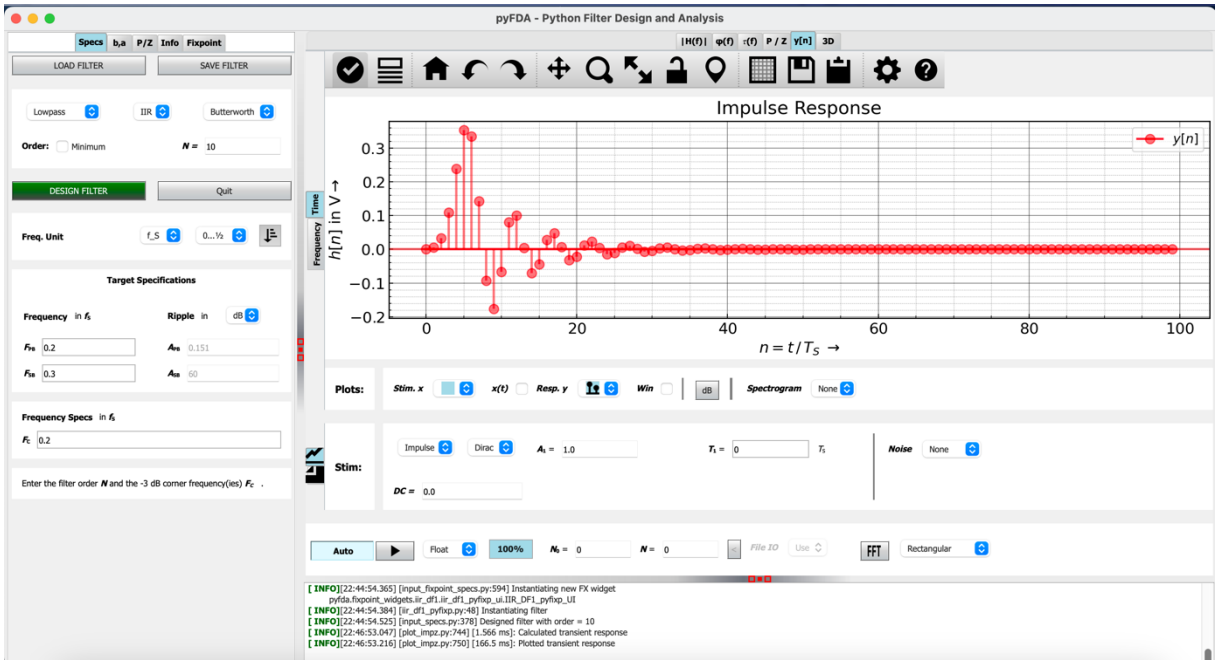
[INFO
[INFO
[INFO

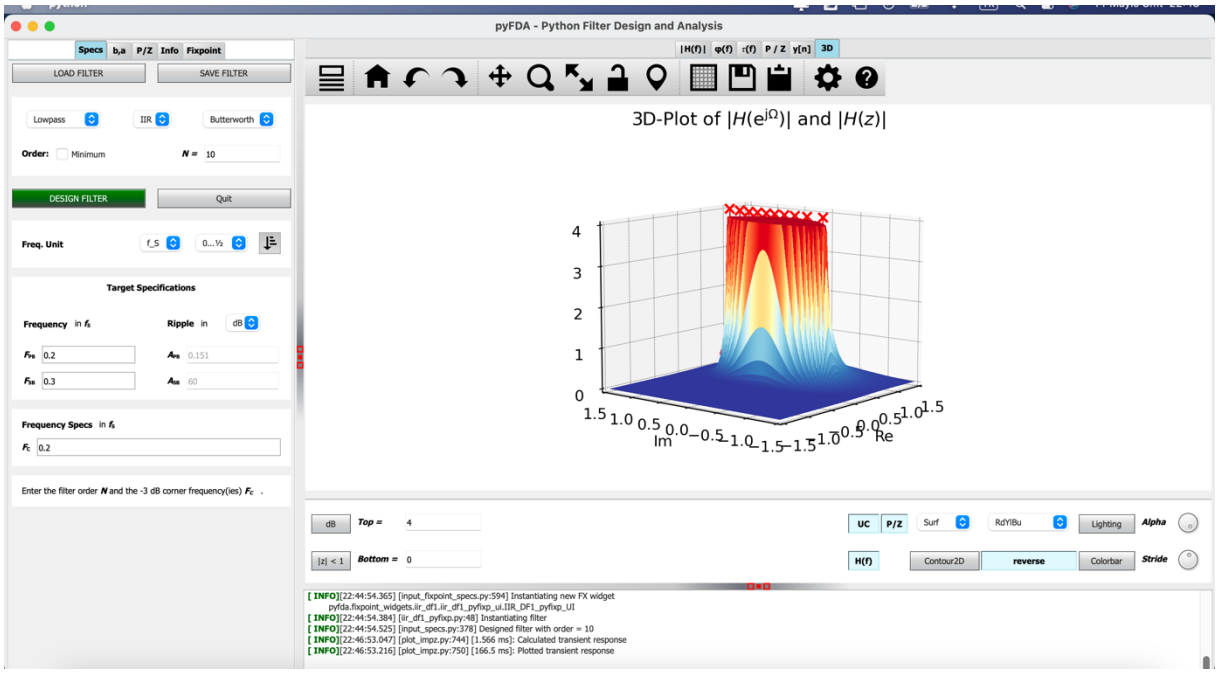
Başka bir filtre tasarlayalım ve dışarıya aktaralım



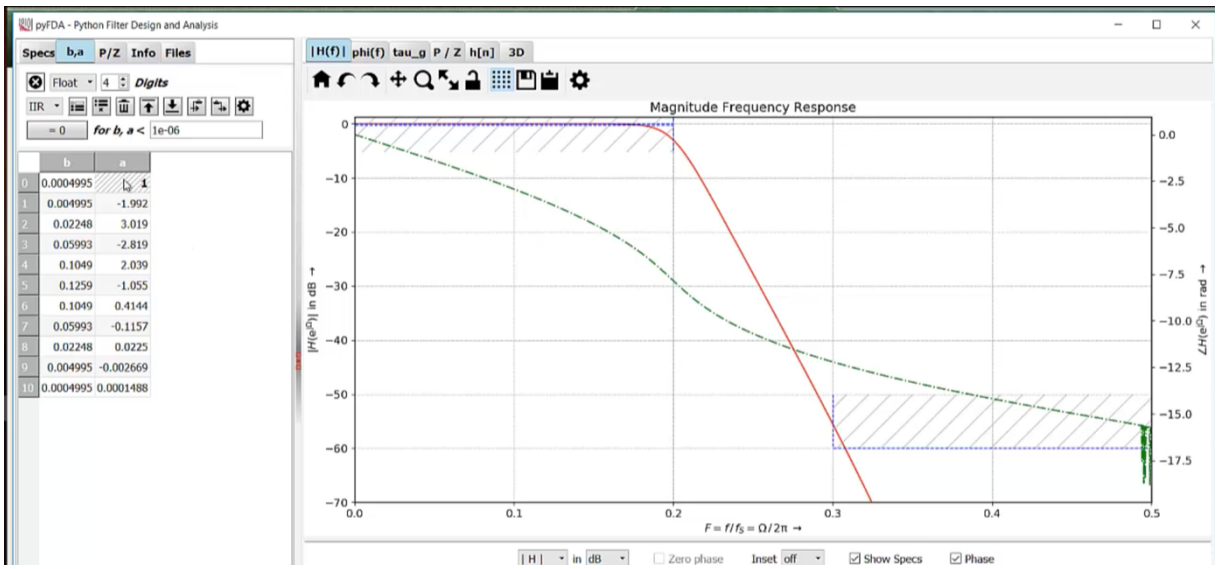



pyFDA - Python Filter Design and Analysis
Specs
b,a
P/Z
Info
Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
IIR
Butterworth
Order:
Minimum
N =
10
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_PB
0.2
A_PB
0.151
F_SB
0.3
A_SB
60
Frequency Specs in f_S
F_C
0.2
Enter the filter order N and the -3 dB corner frequency(ies) F_C .
|H(f)|
φ(f)
τ(f)
P / Z
y[n]
3D
Pole / Zero Plot
Imaginary axis
Real axis
(10)
Overlay:
None
[INFO][22:44:54.344] [input_specs.py:342] Start filter design using method 'Butter.LPman'
[INFO][22:44:54.365] [input_fixpoint_specs.py:594] Instantiating new FX widget pyfda.fixpoint_widgets.iir_df1.iir_df1_pyfixp_ui.IIR_DF1_pyfixp_UI
[INFO][22:44:54.384] [iir_df1_pyfixp.py:48] Instantiating filter
[INFO][22:44:54.525] [input_specs.py:378] Designed filter with order = 10


pyFDA - Python Filter Design and Analysis
Specs
b,a
P/Z
Info
Fixpoint
LOAD FILTER
SAVE FILTER
Lowpass
IIR
Butterworth
Order:
Minimum
N =
10
DESIGN FILTER
Quit
Freq. Unit
f_S
0...½
Target Specifications
Frequency in f_S
Ripple in
dB
F_PB
0.2
A_PB
0.151
F_SB
0.3
A_SB
60
Frequency Specs in f_S
F_C
0.2
Enter the filter order N and the -3 dB corner frequency(ies) F_C .
|H(f)|
φ(f)
τ(f)
P / Z
y[n]
3D
Impulse Response
y[n]
h[n] in V →
n = t / T_S →
Time
Frequency
Plots:
Stim. x
x(t)
Resp. y
Win
dB
Spectrogram
None
Stim:
Impulse
Dirac
A_1 =
1.0
T_1 =
0
T_S
Noise
None
DC =
0.0
Auto
Float
100%
N_S =
0
N =
0
File IO
Use
FFT
Rectangular
[INFO][22:44:54.365] [input_fixpoint_specs.py:594] Instantiating new FX widget pyfda.fixpoint_widgets.iir_df1.iir_df1_pyfixp_ui.IIR_DF1_pyfixp_UI
[INFO][22:44:54.384] [iir_df1_pyfixp.py:48] Instantiating filter
[INFO][22:44:54.525] [input_specs.py:378] Designed filter with order = 10
[INFO][22:46:53.047] [plot_impz.py:744] [1.566 ms]: Calculated transient response
[INFO][22:46:53.216] [plot_impz.py:750] [166.5 ms]: Plotted transient response

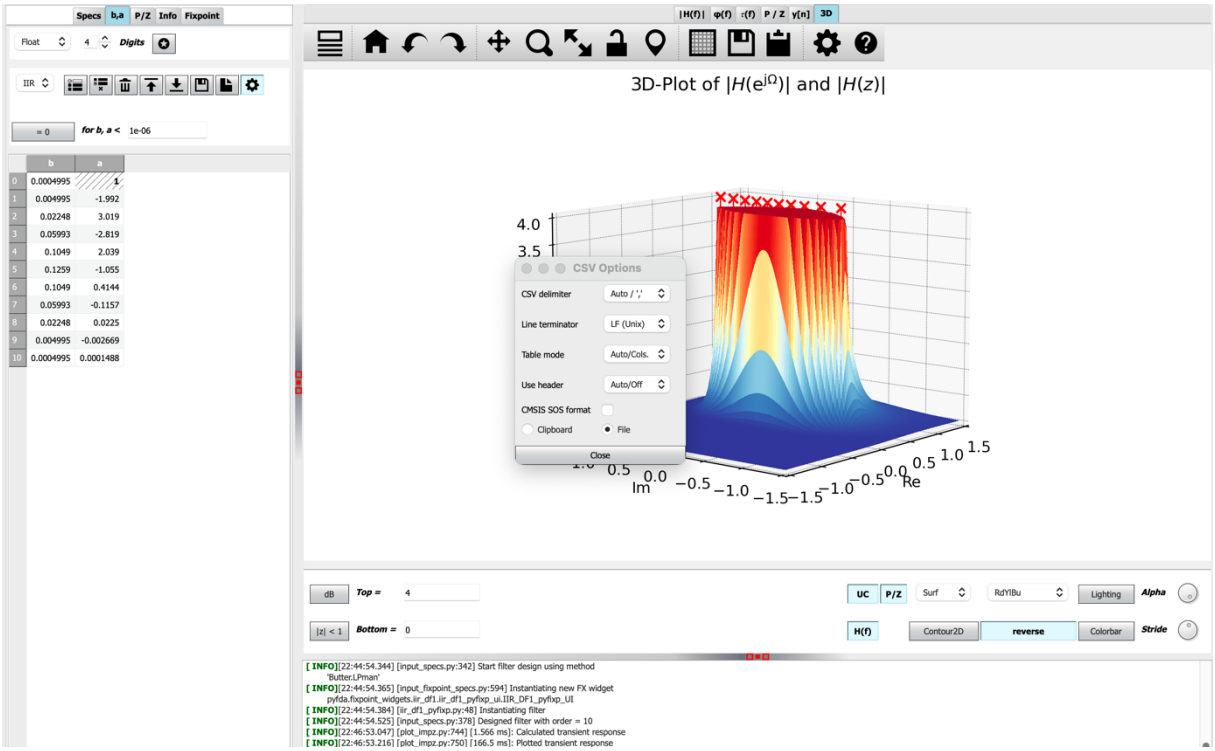


Filtrenin katsayıları

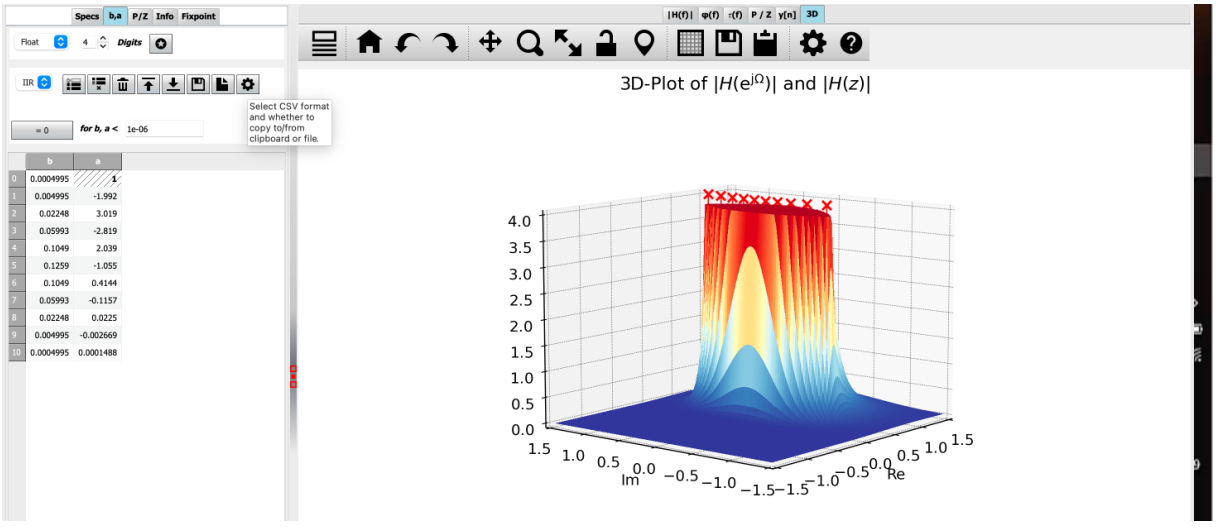


7 Video 7

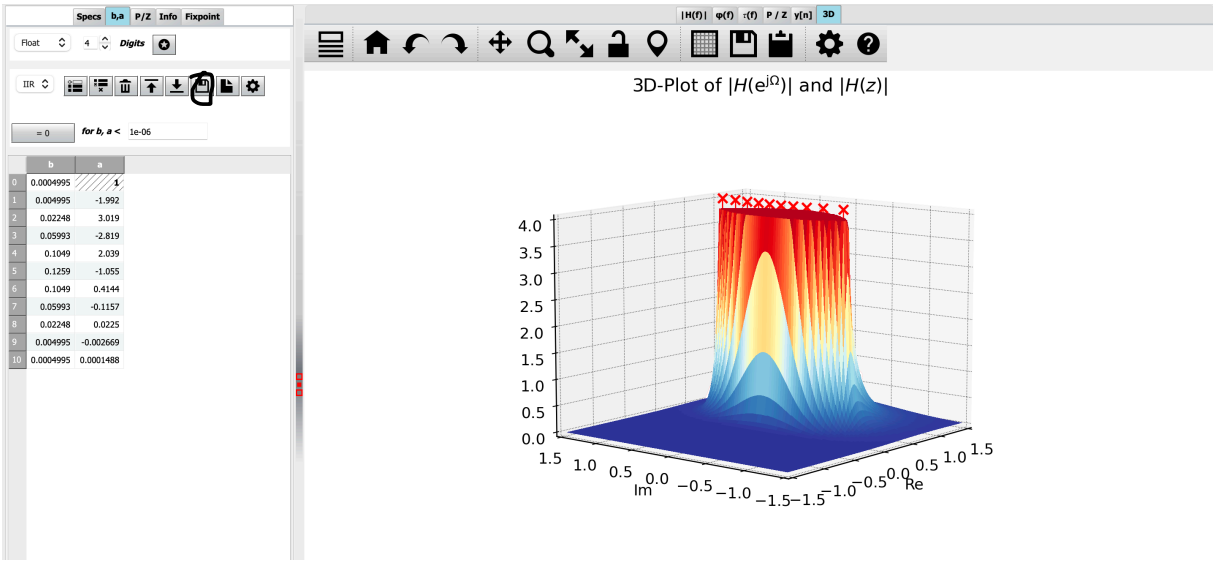
Filtreyi, dışarıya aktarmak için çarka tıklıyoruz

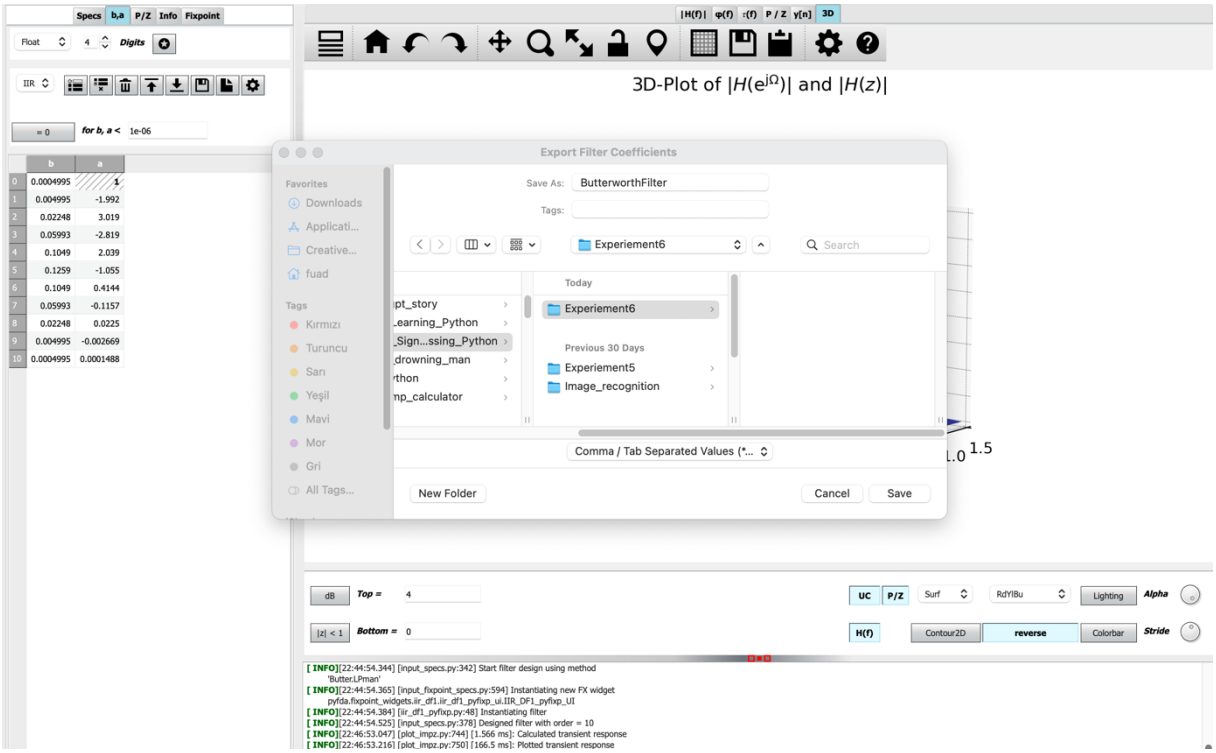


File seçip kapatıyoruz

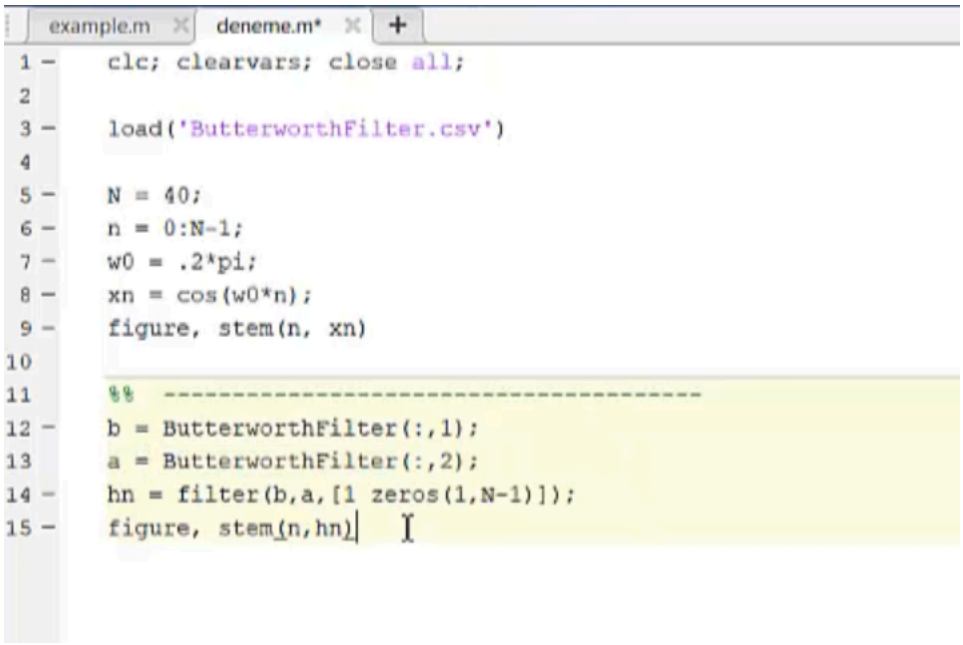


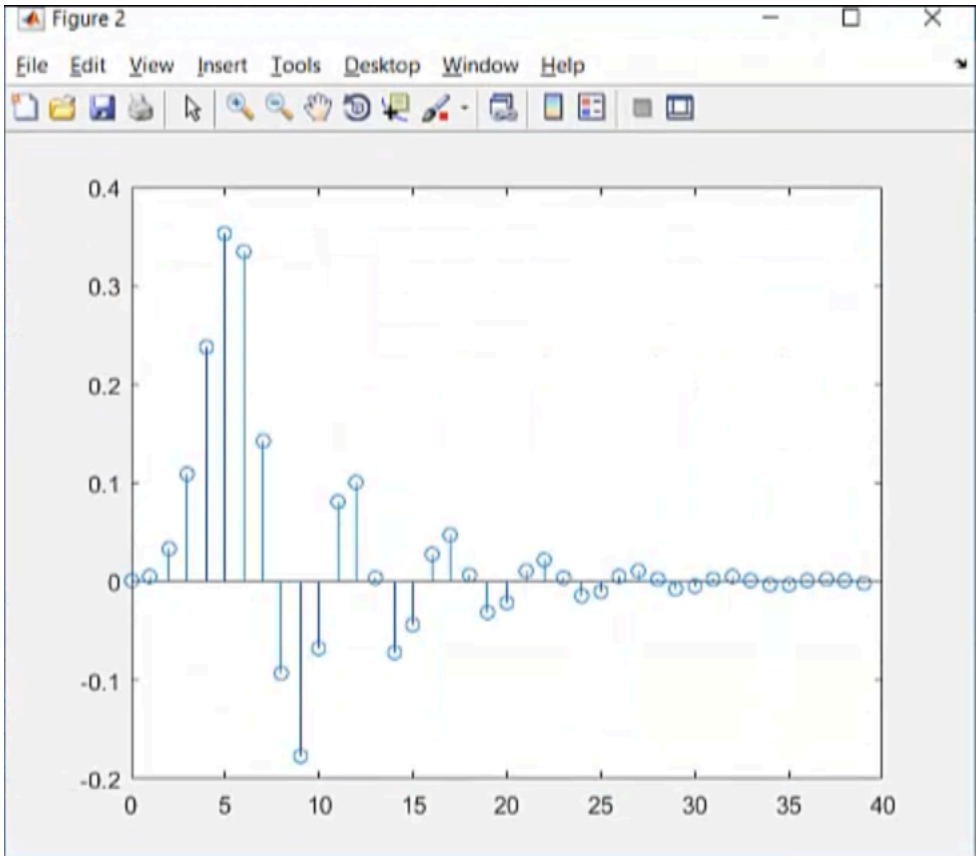
Kaydet tuşuna basılır





Matlab filtre yükleme ve dürtü cevabı





example.m | deneme.m

```
2
3 -   load('ButterworthFilter.csv')
4
5 -   N = 40;
6 -   n = 0:N-1;
7 -   w0 = .2*pi;
8 -   xn = cos(w0*n);
9 -   figure, stem(n, xn)
10
11    %% ---------------------------------------
12 -   b = ButterworthFilter(:,1);
13 -   a = ButterworthFilter(:,2);
14 -   hn = filter(b,a,[1 zeros(1,N-1)]);
15 -   figure, stem(n,hn)
16
17 -   Hw = fftshift(fft(hn));
18 -   Hw_mag = abs(Hw);
19 -   Hw_phs = angle(Hw);
20 -   figure
21 -   subplot 211, plot(Hw_mag)
22 -   subplot 211, plot(Hw_phs)
```

Command Window

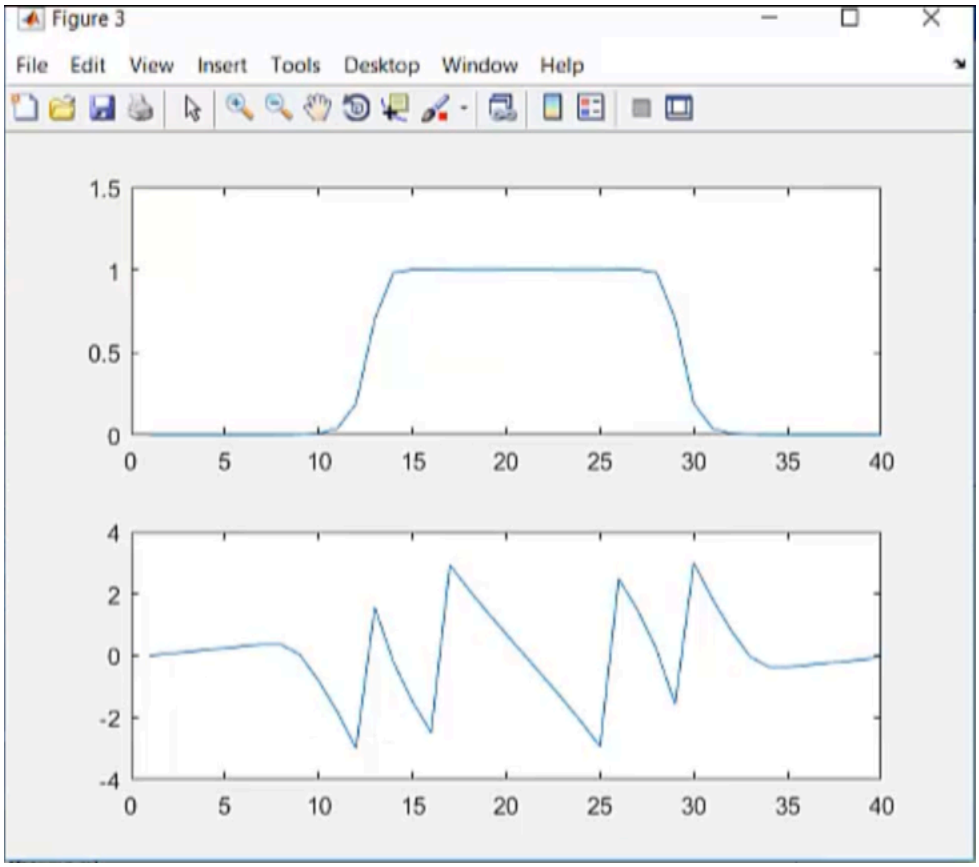
fx >>

```
Hw_phs = angle(Hw);
% figure
% subplot 211, plot(Hw_mag)
% subplot 211, plot(Hw_phs)
tau_w = -diff(Hw_phs)*(N/(2*pi));
```

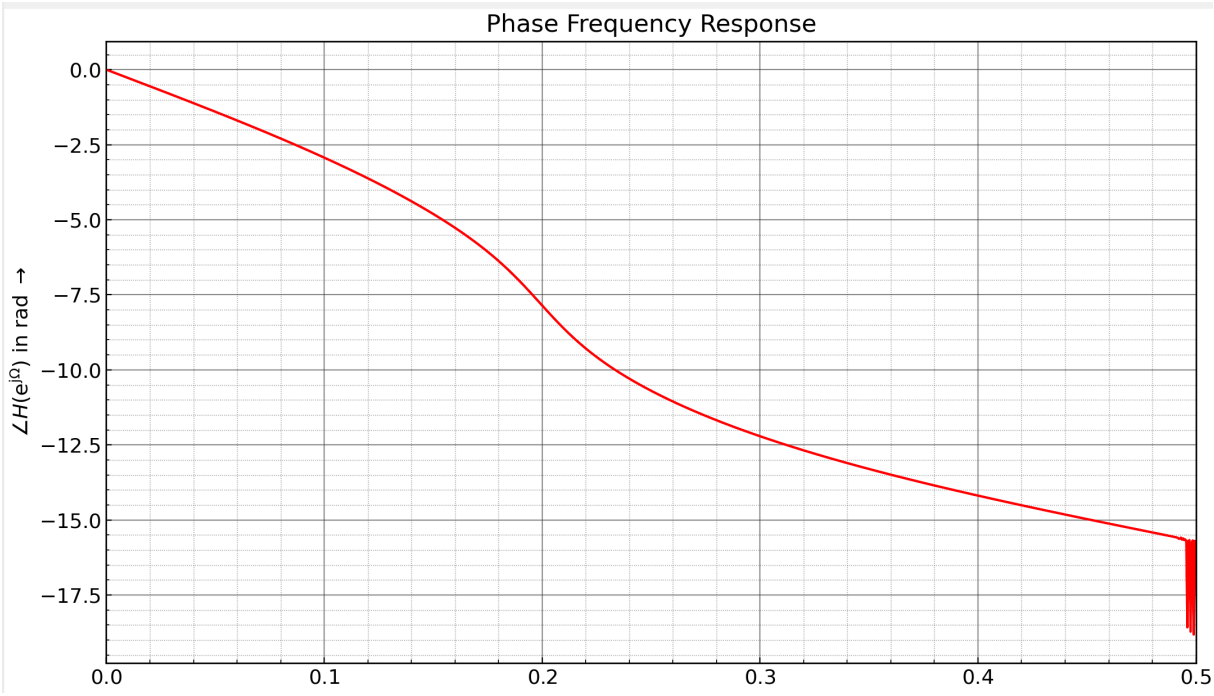
```
tau_w = -diff(Hw_phs)*(N/(2*pi));
figure, plot(tau_w)
```

```
%%  ------------------------------------
b = ButterworthFilter(:,1);
a = ButterworthFilter(:,2);
hn = filter(b,a,[1 zeros(1,N-1)]);
figure, stem(n,hn)

w = linspace(-pi, pi-2*pi/N, N);
Hw = fftshift(fft(hn));
Hw_mag = abs(Hw);
Hw_phs = angle(Hw);
figure
subplot 211, plot(Hw_mag)
subplot 212, plot(Hw_phs)
tau_w = -diff(Hw_phs)*(N/(2*pi));
figure, plot(tau_w)
```



Ama normalde faz cevabı böyle:



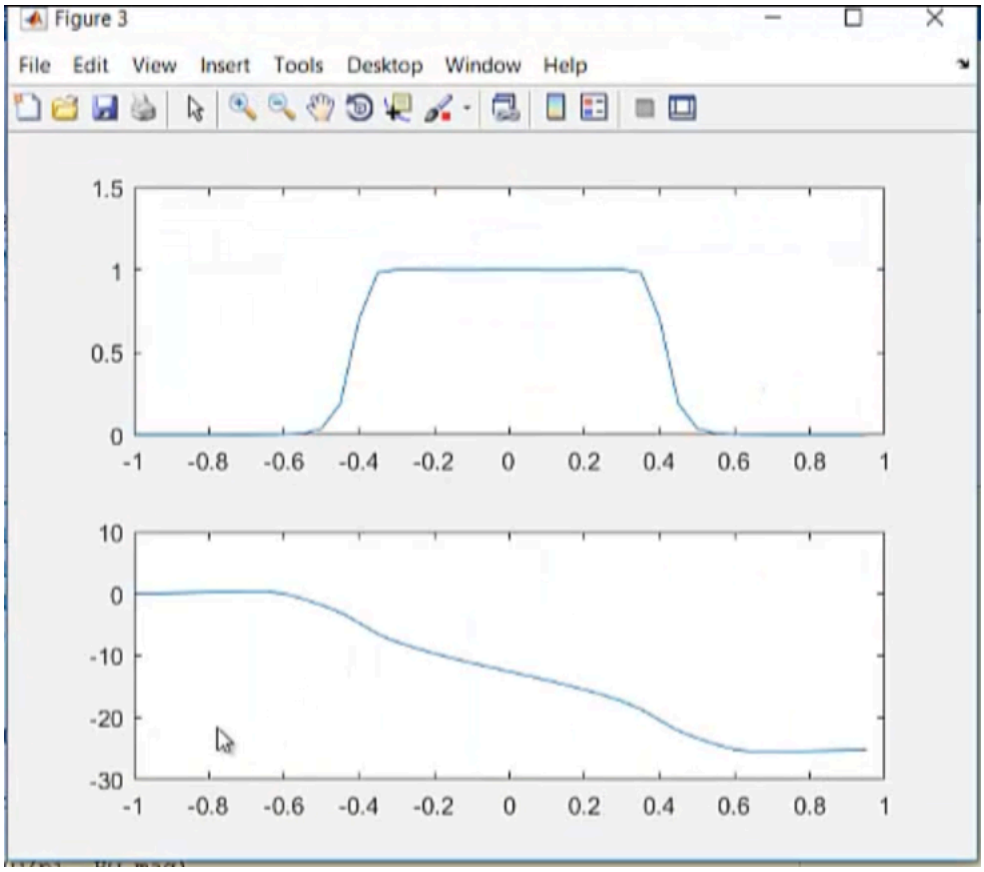
Matlab faz cevabı eksi pi artı pi arasında çizdirmesinden kaynaklıdır. Bunun olmaması için unwrap fonksiyonu kullanılmaktadır.
X eksenini de düzeltelim. Yeni kod:

example.m | deneme.m*

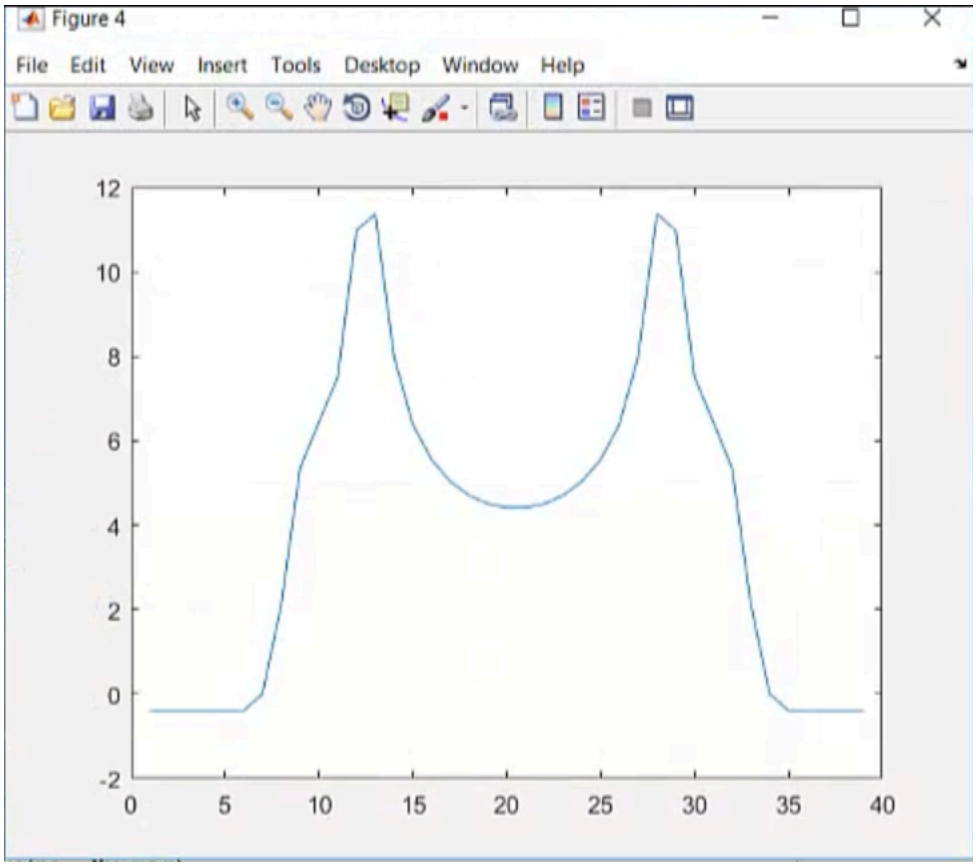
```
n = 0:N-1;
w0 = .2*pi;
xn = cos(w0*n);
figure, stem(n, xn)

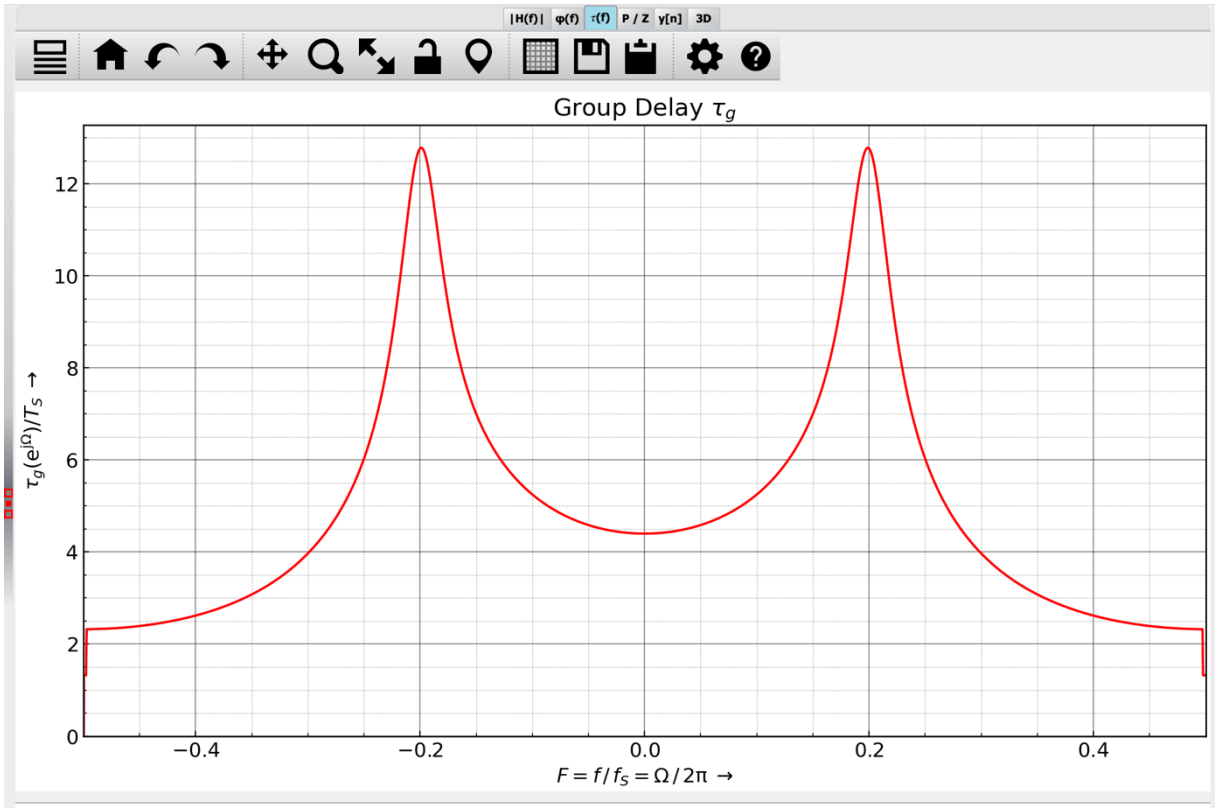
%% ---------------------------------------
b = ButterworthFilter(:,1);
a = ButterworthFilter(:,2);
hn = filter(b,a,[1 zeros(1,N-1)]);
figure, stem(n,hn)

w = linspace(-pi, pi-2*pi/N, N);
Hw = fftshift(fft(hn));
Hw_mag = abs(Hw);
Hw_phs = unwrap(angle(Hw));
figure
subplot 211, plot(w/pi, Hw_mag)
subplot 212, plot(w/pi, Hw_phs)
tau_w = -diff(Hw_phs)*(N/(2*pi));
figure, plot(tau_w)
```



Grup gecikmesi matlab (x ekseni yanlış)


Figure 4
File Edit View Insert Tools Desktop Window Help
12
10
8
6
4
2
0
-2
0 5 10 15 20 25 30 35 40


|H(f)| φ(f) τ(f) P / Z y[n] 3D
Group Delay τ_g
$\tau_g(e^{j\Omega})/T_S \rightarrow$
12
10
8
6
4
2
0
−0.4 −0.2 0.0 0.2 0.4
$F = f/f_S = \Omega/2\pi \rightarrow$

8 Video 8